

# The Kernel Spinor $\Xi_{\text{ker}}(163)$ : Domain Walls, the Hilbert–Hardy Filter, and the Ramanujan–Heegner Index

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## Abstract

Every physical observable belongs to the image of the projection  $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ . What  $\Pi$  cannot transmit — its kernel — has been assumed to be physically inert. This paper shows it is not.

The Quantum Gauge Theory of Time derives the observable boundary  $\mathbb{R}^4$  as the image of a non-invertible rank-5 projection of a bulk Majorana spinor  $\Xi = C\bar{\Xi}$ . The boundary physics is established: admissible modes, fundamental constants, gauge group, Born rule [1, 2, 3]. The present paper derives the structure of  $\ker \Pi$  from the same filter that generates the boundary — and finds that it is organised, addressed, and non-trivially connected to everything the boundary already knows about itself.

The Pincherle filter  $H(z) = P_4(z)/(1 - \alpha z)$  is a meromorphic function on the Riemann sphere  $\mathbb{P}^1$ . Its Hardy–Hilbert decomposition on the unit circle yields the central result of this paper:

$$\Xi_{\text{ker}}(163) = -i \mathcal{H}[\Psi_{\text{boundary}}],$$

the kernel spinor is the Hilbert transform of the boundary spinor. The projection  $\Pi$  is the act of taking the real part of the analytic signal on  $S^1(\tau)$ ; the kernel is its imaginary part; the bulk Majorana spinor  $\Xi = C\bar{\Xi}$  is the complete analytic signal. What is not observed is not absent — it is the imaginary part that makes the real part possible.

The internal structure of the kernel is not imposed from outside. The filter  $H(z) = P_4(z)/(1 - \alpha z)$  has two factors: the degree-4 polynomial  $P_4$  contributes the scale hierarchy  $S_5$ ; the pole at  $z = 1/\alpha$  contributes the tensor sector  $T_8$ . The internal branch  $\mathcal{B}_{13} = S_5 \oplus T_8$  has dimension  $13 = 5 + 8$ . Both components are forced by the rank-5 projection structure:  $S_5$  from the five logarithmic SVD modes [3],  $T_8$  from the anisotropic lock-in tensor  $(\xi, \eta)$  at  $\xi^*/\eta^* = 11/5$  (derived in Section 3 of this paper).

The structure closes arithmetically. The bulk spinor carries Clifford spectral curvature  $\sqrt{5}$ ; the kernel spinor carries modular signature  $R(e^{-2\pi}) = 1/\varphi$  at the lock-in point  $\tau = i$ . These are not two independent numbers: the identity  $\varphi + R(e^{-2\pi}) = \sqrt{5}$  is the closure relation of  $\mathbb{Q}(\sqrt{5})$ . The projection separates  $\varphi$  from  $1/\varphi$ ; the Hilbert transform exchanges them; the CM theory of  $\mathbb{Q}(\sqrt{-163})$  collects them into the largest Heegner number 163, the unique

discriminant of class number 1 that simultaneously contains  $\sigma = 137$  as a Gaussian prime and  $26 = 2 \times 13$  as the kernel internal dimension. Both are forced; 163 follows.

The residual  $\delta_{163} \approx 2.796 \times 10^{-3}$  measures how far the cycle is from closing exactly — a closure forbidden by  $\pi \notin \mathbb{Q}(\sqrt{5})$  (Lindemann–Weierstrass). It lives in the window  $e/10^3 < \delta_{163} < 2\sqrt{2}/10^3$ , bounded below by the entropic scale of the projection and above by the spinorial breathing attractor  $\sqrt{B^*} = 2\sqrt{2}$ . The observed CMB temperature satisfies the same window:  $T_{\text{CMB}}/10^3 \approx 2.7255 \times 10^{-3} \in (e/10^3, 2\sqrt{2}/10^3)$ . The modular residual and  $T_{\text{CMB}}$  are two readouts of the same decoherent  $n = 0$  floor: arithmetic in the kernel, thermal on the boundary. The structural derivation of  $T_{\text{CMB}}$  from the floor dynamics is carried out in Appendix B of this paper. The non-closure  $\delta_{163} \neq 0$  is protected by the same transcendence  $\pi \notin \mathbb{Q}(\sqrt{5})$  that prevents arithmetic closure.

The dynamical invariance  $L_*(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$  and its analytic reformulation at  $\tau_{163}$  are theorems proved in Appendix A. The Noether-norm derivation of  $\lambda_{\text{lock}}$  closes the architecture.

*Epistemic status.* Proposition 6.1 ( $\Xi_{\text{ker}} = -i\mathcal{H}[\Psi_{\text{bdy}}]$ ), Theorems 7.3–7.5 (dynamical and analytic closure), Proposition 7.6 (Noether-norm derivation of  $\lambda_{\text{lock}}$ ), and the Heegner index decomposition  $163 = 137 + 26$  are proved results. The identification of  $T_{\text{CMB}}$  with the arithmetic window is a structural observation: the derivation of  $T_{\text{CMB}}$  from the  $n = 0$  floor is carried out in Appendix B; the quantitative harmonic realisation of  $C_\ell$  on the CMB sky goes beyond the scope of this paper.

## 1 Introduction

Every consistent theory of observation has a kernel. The question has always been: what is in it?

In standard quantum mechanics, the kernel of the measurement process is taken to be empty by convention — the Born rule is postulated to exhaust all physical content. In gauge theories, the kernel of the field equations (the gauge orbits) is removed by fixing, not studied. In string theory, compactification sends the extra dimensions into a small manifold whose internal structure is engineered, not derived. In all three cases the answer is either “nothing physical” or “whatever we need to put there.” That the observables of the Standard Model and General Relativity cannot be derived from within those theories as necessary consequences of internal consistency is a structural limitation, not an accident; it motivates the present framework [2].

The Quantum Gauge Theory of Time (QGT) arrives at the same question from a different direction [1, 2]. The observable boundary  $\mathbb{R}^4$  is the image of a non-invertible projection  $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$  of a single bulk Majorana spinor  $\Xi = C\bar{\Xi}$ . The kernel  $\ker \Pi \neq \{0\}$  is not a gauge artifact. It is the set of bulk configurations that project to the same boundary image — configurations the boundary cannot distinguish, but which are physically real in the bulk. Everything the boundary sees is established: the admissible modes  $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ , the fine-structure constant  $\alpha^{-1} = 20\varphi^4$ , the boundary algebra as quantum mechanics with Born rule derived from projection geometry, and the Wheeler–DeWitt obstruction resolved by the timeless bulk constraint [1, 2, 3, 4]. What the boundary sees is precisely characterised.

This paper asks what the boundary *does not* see — and finds that this too is precisely characterised.

The invisible is not empty. It is organised.

The kernel is organised. It has a specific location — the domain walls  $DW_{1,2}$  where the evanescent edge modes of [3] and [4] live. It has a specific internal dimension —  $13 = 5 + 8$  per chiral branch, assembled from structures already closed in the boundary theory. It has a specific arithmetic index — the largest Heegner number  $163 = 137 + 2 \times 13$ , where  $137 = \sigma$  is the projection residue and 13 is the same integer that appears in  $\sin^2 \theta_W = 3/13$ . And it has a specific modular grammar — the Ramanujan–Heegner layer, in which the Rogers–Ramanujan continued fraction  $R(e^{-2\pi}) = 1/\varphi$  and the  $j$ -invariant  $j(\tau_{163}) = -640320^3$  are the modular readouts of the kernel on the boundary.

The central object of this paper is the kernel spinor:

$$\Xi_{\text{ker}}(163) \in \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+ \subset \ker \Pi, \quad \Xi_{\text{ker}} = C\overline{\Xi_{\text{ker}}}. \quad (1)$$

It is Majorana, because the Majorana condition holds for the full bulk spinor and is inherited by every subspace. It is not observable from the boundary, because it lives in  $\ker \Pi$ . But it is not nothing: its internal structure, its arithmetic index, and its modular signature are all derivable from the same axioms that close the boundary physics.

The paper is structured as follows. Section 2 establishes the domain walls as physical objects with internal structure, beyond the role of boundary conditions.

*Logical position.* This paper is a companion to [1, 2, 3, 4]. The rank-5 projection, its filter  $H(z)$ , the admissible mode set  $\mathcal{N}_{\text{adm}}$ , and the golden SVD hierarchy are established in [2, 3]. The boundary algebra — Born rule, time emergence, Noether invariant  $\mathcal{K}_\tau = \alpha^{-1}$  — is developed in [4]. The present paper takes those results as inputs and derives the structure of the complementary sector  $\ker \Pi$ .

## 2 Domain walls as physical objects

*Every surface that separates what is observed from what is not carries physical content. Not on one side or the other — at the surface itself. The domain walls are where the projection fails to be invertible: where the bulk runs out of room to become boundary. That failure is not an absence. It is the first layer of structure.*

The boundary  $\mathbb{R}^4$  is not uniform. The projection  $\Pi$  selects, from the two-sphere  $S^2$  swept by the Majorana helix on the temporal fibre, a slab between two latitudinal circles. These circles are the domain walls:

$$DW_1 : \theta = \frac{\pi}{2}, \quad n = 2, \quad k = 4, \quad (2)$$

$$DW_2 : \theta = \frac{\pi}{11}, \quad n = 11, \quad k = 1. \quad (3)$$

$DW_1$  at the equator separates the two chiral hemispheres  $\Psi^+$  and  $\Psi^-$ .  $DW_2$  near the north pole is the upper boundary of the observable slab, beyond which the mode  $n = 11$  enters  $\ker \Pi$ . The lock-in at  $n = 5$  sits between them, at  $\theta = \pi/5$ .

The domain walls are the location of evanescent edge modes. The boundary condition  $\partial_n \Psi|_{\text{edge}} = 0$  holds at each wall, and the mode profile decays as  $|\Psi(z)| \sim e^{-z/\ell_{\text{pen}}}$  into the bulk. The penetration depth is derived from the Noether condition at the boundary (Section 2.1):

$$\ell_{\text{pen}} = \frac{1}{\pi \alpha \Gamma_\tau}. \quad (4)$$

The edge modes sustain the exchange oscillation  $\dot{\xi} = -\Gamma_\tau \eta$ ,  $\dot{\eta} = +\Gamma_\tau \xi$  on the invariant surface  $\Sigma : \xi \eta = \alpha^{-1}$  [3] without transporting information across the wall. The normal flux vanishes; the tangential coherence is maintained.

This paper takes the domain walls as the primary objects, not as boundary conditions. The question is: what is the internal structure that the domain walls sustain in the kernel?

### The kernel is localised at the domain walls

The kernel  $\ker \Pi$  consists of all bulk configurations  $\Xi_{\ker}$  such that  $\Pi \Xi_{\ker} = 0$ . Not all of these are physically relevant. The relevant sector is the one that couples to the boundary through the evanescent edge modes — the part of the kernel that the boundary can almost-see, that leaves residue traces in the boundary physics without being itself observable.

This is the kernel at the domain walls: the sector of  $\ker \Pi$  localised within one penetration depth  $\ell_{\text{pen}}$  of  $DW_{1,2}$ . We call it  $\ker \Pi^{\text{loc}}$ . Its characteristic time scale is  $1/\Gamma_\tau = \sqrt{5}/\varphi$ , the inverse of the temporal connection.

## 2.1 Penetration depth from the Noether condition

The penetration depth  $\ell_{\text{pen}}$  is determined by the balance between the evanescent decay in the normal direction  $n$  and the tangential coherence sustained by the Noether invariant. The edge mode  $\Psi_{\text{edge}}(n) \sim \varphi \operatorname{sech}(n/\ell_{\text{pen}})$  satisfies  $\partial_n \Psi_{\text{edge}}|_{n=0} = 0$  and decays as  $e^{-n/\ell_{\text{pen}}}$  for  $n \gg \ell_{\text{pen}}$ . The Noether condition  $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$  on  $\Sigma$  together with the constraint that the edge mode be an admissible mode of the filter  $H(z)$  at the saturation level  $n_{\text{sat}} = 11$  requires:

$$\frac{1}{\ell_{\text{pen}}} = \pi\alpha\Gamma_\tau, \quad \text{i.e.} \quad \ell_{\text{pen}} = \frac{1}{\pi\alpha\Gamma_\tau}. \quad (5)$$

To see this: the filter pole at  $z_{\text{pole}} = \alpha^{-1}$  sets the spectral cutoff, and the temporal connection  $\Gamma_\tau = \varphi/\sqrt{5}$  converts the spectral cutoff to a spatial decay rate. The factor  $\pi$  is the half-cycle of  $S^1(\tau)$ , the minimal observable phase quantum. The product  $\pi\alpha\Gamma_\tau = \pi(\varphi/\sqrt{5})/\alpha^{-1}$  is dimensionless in natural units, confirming that  $\ell_{\text{pen}}$  carries the correct dimension of inverse frequency.

The kernel spinor  $\Xi_{\text{ker}}(163)$  lives in  $\ker \Pi^{\text{loc}}$ : it is Majorana-self-conjugate, evanescent in the normal direction, oscillatory in the tangential direction, and structured in the internal direction by the decomposition of Section 3.

## 3 The internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$

*The boundary knows two fundamental constants:  $\alpha^{-1} = 137$  from the longitudinal channel,  $\sin^2 \theta_W = 3/13$  from the transverse. The kernel holds one number:  $163 = 137 + 2 \times 13$ . Both constants are already inside it, folded together by the same filter that generated them. This section derives how.*

The internal structure of the kernel at the domain walls is not arbitrary. It is constrained by the same rank-5 projection that generates the boundary. We show that the natural internal space has dimension  $13 = 5 + 8$ , assembled from two components that are already closed in the boundary theory.

### The scale component $S_5$

The boundary physics is built on the five logarithmic SVD modes of the Pincherle filter [3]:

$$S_5 = \operatorname{span}\{v_i\}_{i=0}^4, \quad v_i(x) \propto (\log x)^i, \quad (6)$$

with singular values  $\sigma_i \propto \varphi^{-i}$  forming the golden hierarchy. These five modes span the passband of the rank-5 filter: they are what  $\Pi$  carries across to  $\mathbb{R}^4$ .

The same five modes appear in the kernel, but with a different role. On the boundary, the  $v_i$  are the visible output of the filter. Inside the domain walls, they are the scale skeleton of the internal structure: they encode how the kernel is organised across the five cascade levels  $k = 0, \dots, 4$ . The kernel “knows” the scale structure of the boundary because it was generated by the same filter.

This is the first component:  $S_5$ , five-dimensional, established in [3].

### The tensor component $T_8$

The second component of  $\mathcal{B}_{13}$  is the tensor sector of the lock-in field  $(\xi, \eta)$ . At the Noether fixed point, the two chiral amplitudes satisfy  $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$  and the anisotropic ratio  $\xi^*/\eta^* = n_{\text{sat}}/n_* = 11/5$  [3].

The conservation law  $\mathcal{K}_\tau = \alpha^{-1}$  decomposes the projection constant  $\alpha$  into two chiral projections:

$$\alpha_{\parallel} := \alpha \frac{\xi}{\sqrt{\mathcal{K}_\tau}} = \frac{\alpha\xi}{\sqrt{\alpha^{-1}}} = \xi\sqrt{\alpha^3}, \quad \alpha_{\perp} := \alpha \frac{\eta}{\sqrt{\mathcal{K}_\tau}} = \eta\sqrt{\alpha^3}. \quad (7)$$

At the lock-in  $\xi^*/\eta^* = 11/5$ , these two components are linearly independent:  $\alpha_{\parallel}/\alpha_{\perp} = 11/5 \neq 1$ . They span a two-dimensional coupling space at the domain wall.

The exchange equations  $\dot{\xi} = -\Gamma_{\tau}\eta$ ,  $\dot{\eta} = +\Gamma_{\tau}\xi$  on  $\Sigma : \xi\eta = \alpha^{-1}$  define two natural frequencies:

$$\omega_L = \Gamma_{\tau} \cdot \frac{n_*}{n_{\text{sat}}} = \Gamma_{\tau} \cdot \frac{5}{11}, \quad \omega_T = \Gamma_{\tau} \cdot \frac{n_{\text{sat}}}{n_*} = \Gamma_{\tau} \cdot \frac{11}{5}, \quad (8)$$

satisfying  $\omega_L\omega_T = \Gamma_{\tau}^2 = \varphi^2/5$  exactly. These are the longitudinal (gravitational) and transverse (electromagnetic) lock-in frequencies forced by the anisotropy  $\xi^*/\eta^* = 11/5$  of the Noether surface. The tensor kernel of the exchange block is:

$$J_{\xi\eta} = \begin{pmatrix} 0 & -\omega_L \\ \omega_T & 0 \end{pmatrix}, \quad \omega_L\omega_T = \Gamma_{\tau}^2 = \frac{\varphi^2}{5}. \quad (9)$$

The rank-5 projection selects three independent boson channels from the bulk gauge sector: the minimally admissible non-Abelian sector at the boundary requires exactly three channels consistent with the rank-5 cutoff  $n_{\text{sat}} = 11$ , the confined mode  $n = 7$ , and the spinorial covering  $n_{\text{min}} = 2$ . Each of the three channels  $(B_1, B_2, B_3)$  carries two chiral components  $(\parallel, \perp)$ , giving six further dimensions. Together with the two coupling projections  $(\alpha_{\parallel}, \alpha_{\perp})$ , the minimal tensor space in which the anisotropic lock-in and its three-channel boson structure are simultaneously representable is:

$$T_8 = \text{span}\{\alpha_{\parallel}, \alpha_{\perp}, B_{1,\parallel}, B_{1,\perp}, B_{2,\parallel}, B_{2,\perp}, B_{3,\parallel}, B_{3,\perp}\}, \quad (10)$$

eight-dimensional: 2 coupling projections +  $3 \times 2$  boson channels.

### The internal branch and its dimension

The internal branch of the kernel at the domain walls is:

$$\mathcal{B}_{13} = S_5 \oplus T_8, \quad \dim \mathcal{B}_{13} = 13. \quad (11)$$

The 13 is not a choice. Both  $S_5$  and  $T_8$  are forced by structures established in the QGT corpus: the rank-5 SVD and golden hierarchy [3] for  $S_5$ , and the anisotropic lock-in tensor with three boson channels (Section 3 of this paper) for  $T_8$ . The number 5 from the filter rank and the number  $8 = 2 + 3 \times 2$  from the tensor skeleton combine to give 13.

The full internal structure of the lock-in soliton is therefore:

$$\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+, \quad \sigma = 137, \quad \dim \tilde{H} = 27. \quad (12)$$

This is in precise agreement with the decomposition proposed in [3] Remark 6:  $\tilde{H} = \text{span}\{\sigma\} \oplus H_{13}^- \oplus H_{13}^+$  with  $\dim \tilde{H} = 1 + 13 + 13 = 27$ . What the present paper adds is the explicit content of each 13: it is  $S_5 \oplus T_8$ , not a collection of abstract eigenvectors.

### Why 13 is not independent, and why $\sin^2 \theta_W = 3/13$

The dimension  $\dim \mathcal{B}_{13} = 13$  has two independent structural readings that coincide. This coincidence is the derivation of the electroweak mixing angle.

**First reading: from the filter.** The filter  $H(z) = P_4(z)/(1-\alpha z)$  forces  $\dim \mathcal{B}_{13} = 5+8 = 13$  by the argument of Section 3: five from the polynomial degree, eight from the tensor structure of the pole. The eight dimensions of  $T_8$  are explicit:

$$T_8 = \text{span}\{\alpha_{\parallel}, \alpha_{\perp}, W_{\parallel}^-, W_{\perp}^-, Z_{\parallel}^0, Z_{\perp}^0, W_{\parallel}^+, W_{\perp}^+\}, \quad \dim T_8 = 2 + 3 \times 2 = 8. \quad (13)$$

The two coupling projections  $(\alpha_{\parallel}, \alpha_{\perp})$  are forced by the anisotropy  $\xi^*/\eta^* = 11/5$  of the Noether surface (Section 3). The three boson pairs  $(W^-, Z^0, W^+)$ , each carrying two chiral components

$(\parallel, \perp)$ , are forced by the following: the pole sector  $(1 - \alpha z)^{-1}$  at the lock-in  $\xi^*/\eta^* = 11/5$  sustains a non-Abelian internal structure. The minimal non-Abelian compact Lie group with a faithful two-dimensional spinorial representation is  $SU(2)$ , which has exactly three generators. Three generators, each with two chiral components, give six dimensions; together with the two coupling projections,  $\dim T_8 = 8$ . The count  $|\mathcal{C}_{\text{obs}}| = 3$  is the number of  $SU(2)$  generators — not a choice, but the rank of the minimal non-Abelian group that the lock-in anisotropy forces.

**Second reading: from the projection counting.** The rank-5 projection with  $|\mathcal{C}_{\text{obs}}| = 3$  observable boson channels and spinorial covering degree  $n_{\text{min}} = 2$  gives:

$$2r + |\mathcal{C}_{\text{obs}}| = 2 \times 5 + 3 = 13 = \dim \mathcal{B}_{13}. \quad (14)$$

The two readings give the same 13 for the same reason: both count the total capacity of the rank-5 tensor field with the minimal non-Abelian boson sector.

**The electroweak mixing angle.** The transverse projection of the three-channel boson sector onto the observable boundary reads:

$$\sin^2 \theta_W = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13}. \quad (15)$$

This is not a fit to data. The denominator 13 is the internal dimension  $\dim \mathcal{B}_{13}$ , already forced by the filter structure. The numerator 3 is the number of  $SU(2)$  generators, already forced by the lock-in anisotropy requiring the minimal non-Abelian group. Both enter from the same rank-5 projection geometry, through two independent paths that agree.

The observable boundary reads the same object in two ways:  $\alpha^{-1}$  from the norm  $\xi\eta$  (the scalar contraction of the coupling tensor), and  $\sin^2 \theta_W = 3/13$  from the anisotropy  $\xi/\eta = 11/5$  (the non-Abelian contraction). Two numbers, one tensor, one rank-5 projection. The kernel carries their common origin as the single integer  $163 = 137 + 26$ .

*Why 163, and not some other integer.* At this point the two key numbers are in hand: the projection residue  $\sigma = 137$  (derived in Appendix B and in [3]) and the kernel internal dimension  $\dim \mathcal{B}_{13} = 13$  (this section). Their sum  $\sigma + 2 \times \dim \mathcal{B}_{13} = 137 + 26 = 163$  is forced to be the arithmetic index of the kernel. The reason 163 is special — and not, say,  $137 + 28$  or  $127 + 26$  — is a theorem in algebraic number theory: 163 is the largest discriminant for which the ring of integers  $\mathcal{O}_{163}$  of  $\mathbb{Q}(\sqrt{-163})$  has class number 1 (Baker–Stark [11, 12]). Class number 1 is exactly the condition that guarantees that  $\sigma = 137$  is a Gaussian prime in  $\mathcal{O}_{163}$ : it cannot decompose further, so it is irreducible as a boundary residue. With  $\sigma$  and  $\dim \mathcal{B}_{13}$  both fixed by the rank-5 projection geometry, the discriminant  $163 = 137 + 26$  is the unique Heegner number of class 1 that accommodates both simultaneously. No choice is made; both numbers were already determined. The modular grammar of Section 5 is the consequence of this arithmetic structure, not its origin. The next section makes that structure explicit.

## 4 The Heegner phasor algebra and the quadratic partition

*The numbers 5, 8, 13, 19, 43, 67, 137, 163 that appear in the kernel structure are not parameters. They are readouts of a single algebraic object: the quadratic norm forms of imaginary quadratic fields of class number one. This section derives the partition  $\mathcal{B}_{13} = S_5 \oplus T_8$  from first principles of algebraic number theory, establishes the origin of the filter pole  $\alpha_{\Pi}^{-1}$  as a Fibonacci quadratic norm, and reads the domain wall algebra from the mod-13 residue structure.*

### 4.1 Nine fields, nine norm forms

The class-number-one theorem of Baker and Stark [11, 12] asserts that there are exactly nine imaginary quadratic fields  $\mathbb{Q}(\sqrt{-d})$  with class number  $h(K_d) = 1$ . Class number one means

unique factorisation in the ring of integers  $\mathcal{O}_{K_d}$ : every ideal factors uniquely, and every element's norm is represented by a single quadratic form with no fragmentation. This arithmetic rigidity is what makes a field's structure *stable as a memory mode*.

The ring of integers of  $\mathbb{Q}(\sqrt{-d})$  is

$$\mathcal{O}_{K_d} = \begin{cases} \mathbb{Z}\left[\frac{1+\sqrt{-d}}{2}\right] & d \equiv 3 \pmod{4}, \\ \mathbb{Z}[\sqrt{-d}] & d \equiv 1, 2 \pmod{4}, \end{cases} \quad (16)$$

with quadratic norm  $N_d(z) = z\bar{z} \in \mathbb{Z}_{\geq 0}$  for  $z \in \mathcal{O}_{K_d}$ .

| $d$ | $\omega_d$ (generator)    | Norm form $N_d(a, b)$ | $N_d^{(2)}$ | Role                            |
|-----|---------------------------|-----------------------|-------------|---------------------------------|
| 1   | $i$                       | $a^2 + b^2$           | 2           | $H_{\text{bdy}}$                |
| 2   | $\sqrt{-2}$               | $a^2 + 2b^2$          | 2           | $H_{\text{bdy}}$                |
| 3   | $\frac{1+\sqrt{-3}}{2}$   | $a^2 + ab + b^2$      | 3           | $H_{\text{bdy}}$                |
| 7   | $\frac{1+\sqrt{-7}}{2}$   | $a^2 + ab + 2b^2$     | 2           | $H_{\text{bdy}}$                |
| 11  | $\frac{1+\sqrt{-11}}{2}$  | $a^2 + ab + 3b^2$     | 3           | $H_{\text{bdy}}$                |
| 19  | $\frac{1+\sqrt{-19}}{2}$  | $a^2 + ab + 5b^2$     | 5           | $H_{\text{ker}}, \text{gen. 1}$ |
| 43  | $\frac{1+\sqrt{-43}}{2}$  | $a^2 + ab + 11b^2$    | 11          | $H_{\text{ker}}, \text{gen. 2}$ |
| 67  | $\frac{1+\sqrt{-67}}{2}$  | $a^2 + ab + 17b^2$    | 17          | $H_{\text{ker}}, \text{gen. 3}$ |
| 163 | $\frac{1+\sqrt{-163}}{2}$ | $a^2 + ab + 41b^2$    | 41          | ribbon closure                  |

Here  $N_d^{(2)}$  is the smallest norm value attained for  $(a, b) \neq (1, 0)$  — the *second distinct norm* after the minimal value  $N_d(1, 0) = 1$ . The partition  $H = H_{\text{bdy}} \cup H_{\text{ker}} \cup \{163\}$  is determined entirely by the spectral sequence of the norm form — not by the discriminant values themselves.

## 4.2 The partition via $N_d^{(2)}$ : phasor vs spinorial depth

Before the algebra, the geometry.

Every class-number-one field  $\mathbb{Q}(\sqrt{-d})$  defines a phasor  $\rho_d \in U(1)$ : a unit complex number built from the minimal non-unit element of its ring of integers. The phasor traces a trajectory on the circle  $S_\chi^1$  as the temporal fibre parameter  $\tau$  advances. The question that determines the partition is simple: *does the trajectory close?*

For the five boundary fields  $d \in \{1, 2, 3, 7, 11\}$ , the answer is yes. The phasor closes in  $m = 2p$  steps, where  $p \in \mathcal{N}_{\text{adm}}$  is the corresponding admissible mode. The closure is exact, cyclotomic, and independent of  $d$ : the second norm  $N_d^{(2)} \in \{2, 3\}$  stays fixed on the two smallest primes regardless of how large  $d$  grows. These phasors *live on the circle*: they perceive only angle, not depth. They are the boundary modes of the QGT — what  $\Pi$  carries across to  $\mathbb{R}^4$ .

For the three kernel fields  $d \in \{19, 43, 67\}$ , the trajectory does not close on  $S_\chi^1$ . The second norm  $N_d^{(2)} \in \{5, 11, 17\}$  grows with  $d$  as  $N_d^{(2)} \approx d/4$  — the phasor acquires spinorial depth, a capacity to perceive distance in the arithmetic of  $\mathcal{O}_{K_d}$  that no cyclotomic rotation can supply. To close its holonomy, the phasor must be *lifted*: the Majorana self-conjugation condition  $\Xi = C\bar{\Xi}$  maps  $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S_\chi^2$ , taking the phasor from the circle to the two-sphere. Each generation opens one level of spinorial depth that the boundary cannot see.

For  $d = 163$ , the second norm  $N_{163}^{(2)} = 41$  exceeds the saturation index  $n_{\text{sat}} = 11$  of the admissible catalogue. No mode in  $\mathcal{N}_{\text{adm}}$  can accommodate a spinorial step of size 41. The trajectory cannot close at any level accessible to the projection  $\Pi$ . This is not a failure of a particular lift: it



is the statement that the arithmetic of  $\mathbb{Q}(\sqrt{-163})$  lies beyond the reach of  $\Pi$  entirely. The field  $d = 163$  is therefore not a generation of the kernel — it is the arithmetic witness that the list of generations is complete. Baker and Stark proved that no further class-number-one field exists beyond  $d = 163$  [11, 12]; the QGT reads this as the closure of the admissible catalogue.

The second norm  $N_d^{(2)}$  is the discriminating quantity:

| Sector      | $S_5$ (boundary) |   |   |   |    | $T_8$ (kernel) |    |    | $\{163\}$ |
|-------------|------------------|---|---|---|----|----------------|----|----|-----------|
| $d$         | 1                | 2 | 3 | 7 | 11 | 19             | 43 | 67 | 163       |
| $N_d^{(2)}$ | 2                | 2 | 3 | 2 | 3  | 5              | 11 | 17 | 41        |

The  $S_5$  fields ( $d \in \{1, 2, 3, 7, 11\}$ ) have  $N_d^{(2)} \in \{2, 3\}$  — fixed on the two smallest primes regardless of  $d$ . The  $T_8$  fields ( $d \in \{19, 43, 67\}$ ) have  $N_d^{(2)} \in \{5, 11, 17\}$ , growing as  $N_d^{(2)} \approx d/4$  with attractor  $2^2 = 4$ .

This is not a quantitative distinction — it is a qualitative one.

The  $S_5$  fields are *pure phasors*: their second norm stays on  $\{2, 3\}$  because each  $\mathbb{Q}(\sqrt{-d})$  for  $d \in H_{\text{bdy}}$  embeds into a cyclotomic field

$$\mathbb{Q}(\sqrt{-d}) \hookrightarrow \mathbb{Q}(\zeta_m), \quad m \in \{4, 8, 6, 14, 22\} = \{2^2, 2^3, 2 \cdot 3, 2 \cdot 7, 2 \cdot 11\}, \quad (17)$$

with holonomy of order  $m = 2p$  or  $m = 2^k$ ,  $p \in \mathcal{N}_{\text{adm}}$ . A pure phasor lives on  $S_\chi^1$ : it perceives only angle (rotation order), not depth (spinorial distance). The norm  $N_d^{(2)} \in \{2, 3\}$  is the smallest non-trivial element of the cyclotomic rotation group, independent of  $d$ .

The  $T_8$  fields are *spinorial*:  $N_d^{(2)} \sim d/4$  grows with  $d$  because these fields cannot close their holonomy at the cyclotomic level. They require a Majorana lift from  $S_\chi^1$  to the spinorial two-sphere  $S_\chi^2$ . The phasor perceives depth; the norm measures it.

**Proposition 4.1** (Algebraic derivation of the  $S_5/T_8$  partition). *The internal branch  $\mathcal{B}_{13} = S_5 \oplus T_8$  is uniquely determined by the class-number-one norm structure:  $S_5$  consists of the five fields with  $N_d^{(2)} \leq r = 5$  (cyclotomic holonomy, pure phasors);  $T_8$  consists of the three fields with  $N_d^{(2)} > r$  and  $N_d^{(2)} \leq n_{\text{sat}} = 11$  (spinorial holonomy, Majorana-liftable within the admissible catalogue). The field  $d = 163$  has  $N_{163}^{(2)} = 41 > n_{\text{sat}}$  and falls outside both sectors: it closes the Baker–Stark list without contributing a fermionic generation.*

*Proof.* The rank-5 projection admits modes  $n \leq n_{\text{sat}} = 11$  [3]. A field's  $N_d^{(2)}$  measures the minimal spinorial step its phasor requires to distinguish from a unit. For  $N_d^{(2)} \leq r = 5$ : the step falls within the polynomial sector  $P_4(z)$ , whose five logarithmic modes span  $S_5$ . The cyclotomic embedding (17) holds precisely for these five fields; their phasors close within  $S_\chi^1$ . For  $5 < N_d^{(2)} \leq 11$ : the step exceeds the polynomial capacity but lies within the admissible catalogue. The phasors of  $d \in \{19, 43, 67\}$  require the Majorana lift  $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S_\chi^2$ ; they generate  $T_8$ . For  $N_{163}^{(2)} = 41 > n_{\text{sat}} = 11$ : the step exceeds the full admissible catalogue; no further spinorial mode can accommodate it. The field  $d = 163$  is the arithmetic witness of saturation.  $\square$

The ontological reading of this partition is precise. The projection  $\Pi$  is not a map chosen to match observations. It is the unique rank-5 self-similar projection forced by the Fibonacci closure condition  $F_r = r$  with  $r = 5$  [2]. The admissible catalogue  $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$  is forced by the same condition. The class-number-one fields of imaginary quadratic type are forced by algebraic number theory — they are what they are, independent of any physical input.

What the QGT discovers is that these two forced structures *fit together exactly*: the fields whose phasors close cyclotomically ( $N_d^{(2)} \leq r$ ) are precisely the fields whose modes are observable



at the boundary; the fields whose phasors require a spinorial lift ( $r < N_d^{(2)} \leq n_{\text{sat}}$ ) are precisely the fields whose modes constitute the internal kernel.

Nothing is adjusted to make this work. The boundary and the kernel are not designed to complement each other. They are the two sectors that the same arithmetic constraint — class number one — distributes across the same projection threshold  $r = 5$ . The partition  $S_5 \oplus T_8$  is not a decomposition chosen after the fact. It is the form that  $\ker \Pi$  must take if the admissible catalogue is  $\mathcal{N}_{\text{adm}}$  and the class-number-one list is what Baker and Stark proved it to be.

### 4.3 Explicit phasor construction for $d \in \{19, 43, 67\}$

The Heegner phasor for each kernel field is built from the minimal non-unit element of  $\mathcal{O}_{K_d}$ , which is the ring generator  $\omega_d = (1 + \sqrt{-d})/2$  itself (for  $d \equiv 3 \pmod{4}$ , as all three satisfy):

**Generation 1** —  $d = 19$ .  $N_{19}(\omega_{19}) = (1 + 19)/4 = 5 = N_{19}^{(2)}$ .

$$z_{19} = \frac{1 + i\sqrt{19}}{2}, \quad N_{19}(z_{19}) = 5, \quad (18)$$

$$\rho_{19} = \frac{z_{19}}{|z_{19}|} = \frac{1 + i\sqrt{19}}{2\sqrt{5}} \in U(1), \quad (19)$$

$$\chi_{19} = \begin{pmatrix} s_{19}(\tau) \\ \bar{s}_{19}(\tau) \end{pmatrix} \in S_\chi^2, \quad s_{19}(\tau) = \rho_{19} e^{im_{19}\tau}. \quad (20)$$

Condensation eigenvalue  $\lambda_{19} \propto N_{19}^{(2)} = 5 = r$ : the first kernel generation opens at the projection rank.

**Generation 2** —  $d = 43$ .  $N_{43}(\omega_{43}) = (1 + 43)/4 = 11 = n_{\text{sat}}$ .

$$z_{43} = \frac{1 + i\sqrt{43}}{2}, \quad N_{43}(z_{43}) = 11, \quad (21)$$

$$\rho_{43} = \frac{1 + i\sqrt{43}}{2\sqrt{11}} \in U(1), \quad \chi_{43} = \begin{pmatrix} s_{43} \\ \bar{s}_{43} \end{pmatrix}. \quad (22)$$

Condensation eigenvalue  $\lambda_{43} \propto N_{43}^{(2)} = 11 = n_{\text{sat}}$ : the second generation saturates the admissible catalogue.

**Generation 3** —  $d = 67$ .  $N_{67}(\omega_{67}) = (1 + 67)/4 = 17$ .

$$z_{67} = \frac{1 + i\sqrt{67}}{2}, \quad N_{67}(z_{67}) = 17, \quad (23)$$

$$\rho_{67} = \frac{1 + i\sqrt{67}}{2\sqrt{17}} \in U(1), \quad \chi_{67} = \begin{pmatrix} s_{67} \\ \bar{s}_{67} \end{pmatrix}. \quad (24)$$

Condensation eigenvalue  $\lambda_{67} \propto N_{67}^{(2)} = 17$ : the third generation has no admissible mode partner ( $17 \notin \mathcal{N}_{\text{adm}}$ ), which is the structural signature that it is the outermost generation.

The three phasors  $\rho_{19}$ ,  $\rho_{43}$ ,  $\rho_{67}$  generate the three spinorial columns of  $T_8$ . Their condensation eigenvalues  $\{5, 11, 17\}$  are the three values of  $N_d^{(2)}$  for  $d \in H_{\text{ker}}$  — neither input nor fit, but read off the norm form of each field.

#### 4.4 Bridge 4: the filter pole as a Fibonacci quadratic norm

The pole of the Pincherle filter at  $z_{\text{pole}} = \alpha_{\Pi}^{-1}$  is not a number chosen to match the fine-structure constant. It is the quadratic norm of the Fibonacci vector  $(F_5, F_4)^\top$  in the metric of  $\mathbb{Z}[\sqrt{5}]$ :

$$\alpha_{\Pi}^{-1} = (5 + 3\sqrt{5})^2 = 20\varphi^4 = c_F^\top G_{\sqrt{5}} c_F, \quad c_F = \begin{pmatrix} F_5 \\ F_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad (25)$$

where  $G_{\sqrt{5}} = \begin{pmatrix} 1 & \sqrt{5} \\ \sqrt{5} & 5 \end{pmatrix}$  is the Gram matrix of  $\mathbb{Z}[\sqrt{5}]$  in the basis  $(1, \sqrt{5})$ . The closure condition  $F_r = r$  (Fibonacci self-similarity of the projection,  $F_5 = 5 = r$ ) selects the vector  $c_F = (F_5, F_4)^\top = (5, 3)^\top$ ; the quadratic norm in  $\mathbb{Q}(\sqrt{5})$  then gives  $(5 + 3\sqrt{5})^2 = 20\varphi^4$ .

The integer part is  $\sigma = \lfloor \alpha_{\Pi}^{-1} \rfloor = 137$ . This is not reverse-engineered from the fine-structure constant: it is the integer part of a Fibonacci quadratic norm, forced by  $F_r = r$  and the golden field  $\mathbb{Q}(\sqrt{5})$ .

The bridge identity that ties the continuous and discrete languages:

$$\varphi + \frac{1}{\varphi} = \sqrt{5}, \quad \text{i.e.} \quad \varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}, \quad (26)$$

where  $\mathcal{R}_{\text{QGT}} := 1/\varphi$  is both the first singular value  $\sigma_1$  of the rank-5 projection [3] and the arithmetic return of the kernel. The boundary reads  $\sqrt{5}$  as the Clifford spectral curvature of the bulk; the kernel returns  $1/\varphi$  as its modular echo. Projection and kernel are two readings of the same quadratic field  $\mathbb{Q}(\sqrt{5})$ .

#### 4.5 The discriminant scaling chain

The nine Heegner discriminants are not scattered. They form a chain determined by three structural constants alone: the saturation index  $n_{\text{sat}} = 11$ , the Majorana breathing quantum  $B^* = 8 = 2^3$ , and the ribbon dimension  $\dim \mathcal{B}_{13} = 13$ .

$$\begin{aligned} n_{\text{sat}} &= 11 && \text{seed: saturation mode of } \mathcal{N}_{\text{adm}} \\ 11 + B^* &= 19 && \text{generation 1: one breathing quantum} \\ 19 + 3B^* &= 43 && \text{generation 2: three EW traversals} \\ 43 + 3B^* &= 67 && \text{generation 3: confirmed period } 3B^* = 24 \\ 67 + n_{\text{conf}} \cdot 2r &\approx 137 \approx \alpha_{\Pi}^{-1} && \text{(boundary pole, integer part)} \\ 137 + 2 \dim \mathcal{B}_{13} &= 163 && \text{ribbon closure (Baker–Stark)} \end{aligned} \quad (27)$$

with  $B^* = 8 = 2^3$  (Majorana holonomy:  $2 \times 4\pi$  spinorial cycle),  $n_{\text{conf}} = 7$  (the confined mode of  $\mathcal{N}_{\text{adm}}$ ),  $r = 5$  (projection rank),  $|\mathcal{C}_{\text{obs}}| = 3$  (observable boson channels),  $3B^* = 24$  (period  $= |\mathcal{C}_{\text{obs}}| \times B^*$ ).

The generating steps  $\{8, 24, 70, 26\}$  are themselves structural: 8 is the minimal Majorana holonomy;  $24 = 3 \times 8$  is the three-channel EW period; the step from generation 3 to the boundary pole is  $137 - 67 = 70 = n_{\text{conf}} \times 2r = 7 \times 10$ , where the confined mode  $n_{\text{conf}} = 7$  and the monodromy norm  $2r = 10$  multiply; the final step  $26 = 2 \times 13 = 2 \dim \mathcal{B}_{13}$  is the Majorana double cover of the ribbon.

Every discriminant value in the chain is a readout of this structure, not an input to it. The chain also explains the mod-24 congruence: all kernel fields satisfy  $d \equiv 3 \pmod{8}$  (norm form with odd middle coefficient, forced by the ring structure), giving combined period  $\text{lcm}(8, 3) = 24$ , hence  $19 \equiv 43 \equiv 67 \equiv 163 \equiv 19 \pmod{24}$ .

## 4.6 The mod-13 algebra at the domain wall

The domain wall  $DW_{1,2}$  is a gauge-selective filter. The mod-13 structure of the kernel closure encodes precisely what the filter passes and what it retains.

The three key values satisfy:

$$163 \equiv 7 \pmod{13}, \quad 137 \equiv 7 \pmod{13}, \quad 26 \equiv 0 \pmod{13}. \quad (28)$$

The residue class 7 is the signature of the confined mode  $n = 7 \in \mathcal{N}_{\text{adm}}$ :  $n = 7$  does not generate the Pincherle dilation lattice on the boundary and lives only in  $\ker \Pi$ . Both the kernel closure 163 and the boundary pole  $\sigma = 137$  carry this signature. Their difference  $26 = 163 - 137$  is exactly zero modulo  $\dim \mathcal{B}_{13} = 13$ : it is *invisible at the boundary*.

The shadow ribbon  $\mathcal{B}_{13}^{\text{shadow}}$  is the unobservable complement of  $\mathcal{B}_{13}$  at the domain wall:

$$\Pi : \mathcal{B}_{13}^{\text{core}} \longrightarrow \mathcal{B}_{13}^{\text{bdy}}, \quad \ker(\Pi|_{\mathcal{B}_{13}}) \cong \mathcal{B}_{13}^{\text{shadow}}, \quad \dim \mathcal{B}_{13}^{\text{shadow}} = 26. \quad (29)$$

Its internal decomposition:

$$26 = 2 \times (5 + 8) = \underbrace{2 \times 5}_{10} + \underbrace{2 \times 8}_{16} = 2 \dim S_5 + 2 \dim T_8. \quad (30)$$

The projection loses exactly one full ribbon. What passes has residue  $\equiv 7 \pmod{13}$  (via the boundary pole  $\sigma$ ); what stays has residue  $\equiv 0 \pmod{13}$  and is invisible to the observable algebra.

This is the complete statement of the domain wall as an algebraic object. The quantitative derivation of what the domain wall *emits* — the edge map  $G_{\text{edge}, \mathcal{H}}^{\eta\xi} : S_5 \rightarrow T_8$  that governs transitions between the boundary and kernel sectors, and in particular the  $\pi^+ \rightarrow \mu^+ + \nu_\mu$  transition where the pion (boundary,  $\in S_5$ ) decays to the muon (kernel,  $\in T_8$ ) with the neutrino carrying the mod-13 invisible charge — belongs to the matter sector and is deferred to a subsequent paper.

## 4.7 The analytic signal as sech/tanh decomposition

The connection between the Heegner phasor algebra and the Hardy–Hilbert decomposition of Section 6 is made concrete by the mother solution.

The bulk Majorana field at the lock-in takes the form:

$$\Xi_{\text{bulk}} = \varphi \operatorname{sech}(\Gamma_\tau t) - i\varphi \tanh(\Gamma_\tau t), \quad (31)$$

where  $\operatorname{sech}(\Gamma_\tau t) = \operatorname{Re} \Xi_{\text{bulk}}$  is the boundary component  $\Psi_{\text{bdy}}$  and  $\tanh(\Gamma_\tau t) = (-1/\varphi) \operatorname{Im} \Xi_{\text{bulk}}$  is proportional to  $\Xi_{\text{ker}}$ . The Hilbert–Cayley identity

$$\mathcal{H}_{\text{Cay}}[\operatorname{sech}(\Gamma_\tau t)] = \tanh(\Gamma_\tau t) \quad (32)$$

is the operator realisation of Proposition 6.1 ( $\Xi_{\text{ker}} = -i\mathcal{H}[\Psi_{\text{bdy}}]$ ) in the explicit basis of the mother solution. The conservation law is:

$$\operatorname{sech}^2(\Gamma_\tau t) + \tanh^2(\Gamma_\tau t) = 1, \quad (33)$$

which is the constant-modulus condition  $|\Xi_{\text{bulk}}|^2 = \varphi^2$ : the total amplitude of the analytic signal is conserved, and the projection  $\Pi$  merely rotates between its real and imaginary parts.

The Heegner phasor algebra now closes on the analytic signal: the phasors  $\rho_d = (1 + i\sqrt{d})/(2\sqrt{N_d^{(2)}})$  are the kernel’s arithmetic skeleton at the CM points  $\tau_d$ , the sech/tanh decomposition is their dynamical realisation on the temporal fibre, and the Hilbert transform is the operator that exchanges the two. Neither can exist without the other:  $\operatorname{sech}^2 + \tanh^2 = 1$  is

the same identity as  $\varphi + 1/\varphi = \sqrt{5}$ , read as an energy conservation law instead of an algebraic closure.

Sections 2 and 3 established the location and the internal structure of the kernel. This section gives the kernel a name and an operator. The kernel spinor  $\Xi_{\text{ker}}(163)$  is not a new field introduced to account for missing physics. It is what the bulk Majorana spinor  $\Xi = C\bar{\Xi}$  becomes when  $\Pi$  acts and cannot transmit it. It is already there. The only question is what it looks like.

## Definition and Majorana structure

The kernel spinor is the component of the full bulk Majorana field  $\Xi = C\bar{\Xi}$  that lives in the local kernel at the domain walls:

$$\begin{aligned}\Xi_{\text{ker}}(163) &:= (\mathbf{1} - \Pi)\Xi|_{\text{ker } \Pi^{\text{loc}}}, \\ \Xi_{\text{ker}}(163) &= C\overline{\Xi_{\text{ker}}(163)}.\end{aligned}\tag{34}$$

The Majorana condition is inherited: since  $\Xi = C\bar{\Xi}$  holds for the full bulk field, and  $\Pi$  commutes with charge conjugation (the boundary is charge-neutral by [2]), the complement  $(\mathbf{1} - \Pi)\Xi$  also satisfies the Majorana condition. The kernel spinor is therefore self-conjugate, globally neutral, and carries no independent charge sector. It is not a Dirac fermion and not a Weyl spinor. It is a Majorana object in the bulk that the boundary projection cannot transmit.

Physically:  $\Xi_{\text{ker}}(163)$  is the part of the bulk that “tries to cross” but is reflected at the domain walls. It does not vanish: it organises itself into the internal structure  $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$  and leaves evanescent traces on both sides of the wall.

## The internal operator $\Gamma_{163}$

To make the internal structure precise, we introduce an intertwiner  $J : \mathcal{B}_{13}^+ \rightarrow \mathcal{B}_{13}^-$  that represents the Majorana conjugation between the two chiral branches:

$$J = C|_{\mathcal{B}_{13}^+ \rightarrow \mathcal{B}_{13}^-}, \quad J^{-1}J = JJ^{-1} = \mathbf{1}_{13}.\tag{35}$$

From  $J$  we construct the internal operator:

$$\Gamma_{163} := \sqrt{163} \begin{pmatrix} 0 & J \\ -J^{-1} & 0 \end{pmatrix} \quad \text{on} \quad K_{163} := \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+.\tag{36}$$

**Proposition 4.2** (Minimal polynomial of  $\Gamma_{163}$ ).  $\Gamma_{163}^2 = -163 \cdot \mathbf{1}_{26}$ . The normalised involution  $A_{163} := i\Gamma_{163}/\sqrt{163}$  satisfies  $A_{163}^2 = \mathbf{1}_{26}$ , and the associated projectors  $\Pi_{163}^\pm = \frac{1}{2}(\mathbf{1} \pm A_{163})$  are well-defined orthogonal projectors on  $K_{163}$ .

*Proof.* Direct computation:  $\Gamma_{163}^2 = 163 \cdot \begin{pmatrix} 0 & J \\ -J^{-1} & 0 \end{pmatrix}^2 = 163 \cdot \begin{pmatrix} -JJ^{-1} & 0 \\ 0 & -J^{-1}J \end{pmatrix} = -163 \cdot \mathbf{1}_{26}$ . The involution follows:  $A_{163}^2 = (i/\sqrt{163})^2 \Gamma_{163}^2 = (-1/163) \cdot (-163 \cdot \mathbf{1}) = \mathbf{1}$ . Idempotency of  $\Pi_{163}^\pm$  is standard.  $\square$

The number 163 enters not as an external input but as the *minimal polynomial* of an operator whose construction is forced by the Majorana structure of the bulk and the bipartition  $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ . It is the first time a Heegner number appears as a derived algebraic invariant of a physical projection operator.

*Pause here.* 163 was a number observed in number theory — the largest Heegner discriminant, known since Gauss, proved by Baker and Stark. It is now a number derived in physics: the minimal polynomial of an operator that the Majorana condition and the rank-5 projection require. It was not put here. It emerged.

## The minimal lock-in background

The kernel spinor lives against a background fixed by the lock-in. The Rogers–Ramanujan continued fraction at  $q = e^{-2\pi}$  gives the exact modular value [6, 7]:

$$R_\star := R(e^{-2\pi}) = \frac{1}{\varphi} = \varphi - 1, \quad (37)$$

from which the algebraic identity follows immediately:

$$\varphi^2 = 2 + R(e^{-2\pi}). \quad (38)$$

This is a closed algebraic fact, not a conjecture: it follows from  $R_\star = 1/\varphi$  and the golden equation  $R_\star^2 + R_\star - 1 = 0$ . The background amplitude of the lock-in soliton is therefore  $A_\star = \varphi^2 = 2 + R_\star$ , the unique value at which the modular seed and the boundary norm coincide.

The minimal lock-in solution compatible with the QGT boundary conditions of [3] (evanescent normal profile,  $\partial_n \Psi_\star|_{\text{edge}} = 0$ ,  $\xi_\star \eta_\star = \alpha^{-1}$ ,  $P_\sigma \Psi_\star = 0$ ) is:

$$\Psi_\star(s, n) = \varphi \operatorname{sech}\left(\frac{n}{\ell_{\text{pen}}}\right) e^{i\omega_{\text{bal}} s} u_M, \quad u_M \in K_{163}, \quad (39)$$

where  $s$  is the tangential coordinate on  $\Sigma : \xi\eta = \alpha^{-1}$ ,  $n$  is the normal coordinate, and  $\omega_{\text{bal}} = 0.5284$  is the canonical balance frequency of [4]. The sech profile satisfies  $\Psi_\star(0) \neq 0$ ,  $\partial_n \Psi_\star(0) = 0$ , and  $\Psi_\star \rightarrow 0$  as  $|n| \rightarrow \infty$ : it is the minimal profile compatible with an evanescent edge mode that does not vanish at the wall. The amplitude  $\varphi = \sqrt{A_\star}$  is fixed by the Rogers–Ramanujan seed  $R_\star = 1/\varphi$ : no free parameter enters.

## The linearised operator $L_\star^{(13)}$

Writing  $\Psi = \Psi_\star + \varepsilon\chi$  and expanding the nonlinear boundary Dirac equation

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi + \lambda_{\text{eff}}|\Psi|^{2/3}\Psi = 0, \quad (40)$$

with  $\lambda_{\text{eff}} = 16\pi^2\varphi^{12}$  and  $\xi\eta = \alpha^{-1}$ , one obtains the linearised operator  $L_\star$ . The block-diagonal structure

$$P_\sigma L_\star P_{\text{dyn}} = 0, \quad P_{\text{dyn}} L_\star P_\sigma = 0, \quad (41)$$

follows from the Mellin-diagonal action of  $L_\star$  on the mode space: the constant mode  $P_\sigma$  (the pole sector  $\sigma = 137$ ) is orthogonal to the dynamic sector  $P_{\text{dyn}} = \mathbf{1} - P_\sigma$  under the Mellin inner product [3], so cross-coupling vanishes identically. This guarantees that  $\sigma = 137$  is protected under linearisation.

On the internal branch  $K_{163} = \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ , the linearised operator decomposes as:

$$L_\star^{(13)} = L_{\text{scale}}^{(5)} \oplus L_{\text{tens}}^{(8)}, \quad (42)$$

where  $L_{\text{scale}}^{(5)} = \text{diag}(\lambda_0, \dots, \lambda_4)$  acts on  $S_5$  through the five golden singular values, and

$$L_{\text{tens}}^{(8)} = J_{\xi\eta} \otimes I_4 + M_{\text{EW}}, \quad (43)$$

with  $J_{\xi\eta}$  given by (9) and  $M_{\text{EW}}$  an  $8 \times 8$  matrix that mixes the three boson channels  $(B_1, B_2, B_3)$  while leaving  $(\alpha_\parallel, \alpha_\perp)$  normative.

The block form on  $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$  is:

$$L_\star^{(163)} = \mathbb{H}_{13} + \frac{\varepsilon_{163}}{\sqrt{163}} \Gamma_{163}, \quad (44)$$

where  $\mathbb{H}_{13} = \text{diag}(H_{13}^-, H_{13}^+)$  is the diagonal part and  $\varepsilon_{163}$  is the modular residue defined in the next section. The commutation condition  $[L_\star^{(163)}, \Gamma_{163}] = 0$  requires  $H_{13}^- J = J H_{13}^+$ , i.e. the two 13-dimensional blocks are conjugate under  $J$ . This is satisfied by the explicit seed  $\Psi_\star$  of Theorem 7.3 (Appendix A, Step 2): the intertwiner  $J_\star u_\star = -i u_\star$  on the resonant subspace is the operator realisation of this conjugation.

## The internal splitting

If  $\lambda_j$  is an eigenvalue of  $H_{13}^-$ , the eigenvalues of the full  $2 \times 2$  block at level  $j$  are:

$$\lambda_j^{(\pm)} = \lambda_j \pm i \varepsilon_{163}. \quad (45)$$

For the balance mode  $\lambda_j = \omega_{\text{bal}} = 0.5284$ :

$$\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028. \quad (46)$$

Since  $\varepsilon_{163} \approx 2.8 \times 10^{-3} \ll \omega_{\text{bal}} = 0.5284$ , the kernel sector does not replace the boundary dynamics: it splits it internally. The domain walls sustain a doublet structure around every boundary mode, with the doublet gap set by the modular residue of the Ramanujan seed.

## 5 The Ramanujan–Heegner layer

*The kernel has a structure. But structure alone is not enough to say where it lives in the landscape of mathematics. This section gives the kernel its address: the arithmetic of imaginary quadratic fields. Four of the five admissible modes of the QGT are Heegner numbers. The fifth — the lock-in  $n = 5$  — is the first prime to break the class-number-1 condition, and it is precisely there that the boundary acquires its observable form. The arithmetic is not decorative. It is the natural algebraic language in which the kernel architecture is encoded.*

### Established facts

The following are numerical and algebraic facts, not conjectures.

**Proposition 5.1** (Heegner–Admissible characterisation). *The admissible mode set  $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$  is characterised by the following Heegner–Baker–Stark criterion:*

$$\mathcal{N}_{\text{adm}} = \{p \text{ prime} : p \leq 11, h(-4p) = 1 \text{ or } p = 5\}, \quad (47)$$

where  $h(-4p)$  is the class number of  $\mathbb{Q}(\sqrt{-p})$ , and the set  $\{p : h(-4p) = 1\} = \{2, 3, 7, 11\}$  is determined by the Baker–Stark theorem (finite enumeration, no free parameter). The mode  $n = 5$  is the unique prime excluded from  $h(-4p) = 1$  but included in  $\mathcal{N}_{\text{adm}}$ : it is the first prime with  $h(-20) = 2$ , and it is precisely the lock-in mode where the boundary acquires its observable form. This characterisation is proved: all steps reduce to the Baker–Stark classification plus the QGT rank-5 selection  $F_r = r \Rightarrow r = 5$ .

**(F1)** The admissible modes satisfy  $\mathcal{N}_{\text{adm}} \cap \mathcal{H} = \{2, 3, 7, 11\}$ , where  $\mathcal{H} = \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$  is the set of Heegner numbers (Baker–Stark theorem). Four of the five admissible modes are Heegner numbers. The fifth,  $n = 5$ , satisfies  $h(-20) = 2$ : it is the first prime to break the class-number-1 condition, and it is the lock-in.

**(F2)** The Rogers–Ramanujan continued fraction at  $q = e^{-2\pi}$  gives  $R(e^{-2\pi}) = 1/\varphi$  exactly [6]. This is the modular value at the fifth root of unity, i.e. at the modulus of the QGT lock-in ( $n = 5$ ,  $\kappa^2 = 5$ ). Setting  $R := R(e^{-2\pi}) = 1/\varphi = \varphi - 1$ , the golden equation  $R^2 + R - 1 = 0$  gives  $R^2 = 1 - R$ , and therefore:

$$\varphi^2 = (1 + R)^2 = 1 + 2R + R^2 = 1 + 2R + (1 - R) = 2 + R. \quad (48)$$

This is an exact algebraic identity, not an approximation:

$$\varphi^2 = 2 + R(e^{-2\pi}). \quad (49)$$

The kernel spinor carries  $R = 1/\varphi$  as its modular signature; the bulk Majorana spinor carries  $\sqrt{5}$  as its Clifford spectral curvature [4]. These are connected by the unique algebraic identity that closes  $\mathbb{Q}(\sqrt{5})$ :

$$\varphi + \frac{1}{\varphi} = \sqrt{5}, \quad (50)$$

i.e.  $\varphi + R(e^{-2\pi}) = \sqrt{5}$ . The projection  $\Pi$  generates both simultaneously: the bulk spectral curvature ( $\sqrt{5}$ ) and the kernel modular signature ( $R = 1/\varphi$ ) are the two sides of the same closure identity. Neither can exist without the other.

**(F3)** The  $j$ -invariant at the CM point  $\tau_{163} = (1 + i\sqrt{163})/2$  is  $j(\tau_{163}) = -640320^3$  (complex multiplication theory). The near-integer identity  $e^{\pi\sqrt{163}} = 640320^3 + 744 - \varepsilon$ ,  $\varepsilon \approx 7.5 \times 10^{-13}$ , is a consequence.

**(F4)** The Heegner seed connects to the Rogers–Ramanujan value through the exact substitution  $\varphi^2 = 2 + R(e^{-2\pi})$ :

$$e^{\pi\sqrt{163}} \approx \varphi^2 \cdot 10^{17} = (2 + R(e^{-2\pi})) \cdot 10^{17}, \quad (51)$$

placing the Heegner seed  $e^{\pi\sqrt{163}}$  and the Rogers–Ramanujan value  $R(e^{-2\pi}) = 1/\varphi$  in the same formula. This is a rescaling, not a derivation of 163 from  $R$ : the identity  $\varphi^2 = 2 + R$  is exact, but the approximation  $e^{\pi\sqrt{163}} \approx \varphi^2 \cdot 10^{17}$  is the content of Corollary 5.3 below.

Two residuals measure the gap precisely. The *logarithmic residual*:

$$\varepsilon_{163} := \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) = \ln\left(\frac{e^{\pi\sqrt{163}}}{\varphi^2 \cdot 10^{17}}\right) \approx 2.7998 \times 10^{-3}, \quad (52)$$

and the *relative residual*:

$$\delta_{163} := \frac{|e^{\pi\sqrt{163}} - \varphi^2 \cdot 10^{17}|}{e^{\pi\sqrt{163}}} \approx 2.7958 \times 10^{-3}. \quad (53)$$

These are distinct objects ( $\varepsilon_{163} = \ln(1 + \delta_{163}/(1 - \delta_{163})) \approx \delta_{163}$  to leading order) but both satisfy the same window:

$$\frac{e}{10^3} < \delta_{163}, \varepsilon_{163} < \frac{2\sqrt{2}}{10^3}. \quad (54)$$

Numerically:  $e/10^3 \approx 2.7183 \times 10^{-3}$ ,  $\delta_{163} \approx 2.7958 \times 10^{-3}$ ,  $\varepsilon_{163} \approx 2.7998 \times 10^{-3}$ ,  $2\sqrt{2}/10^3 \approx 2.8284 \times 10^{-3}$ . The fact that the same window captures both residuals — the multiplicative error and the additive logarithmic error — independently of the representation, is the structural content of the window: it is not an artefact of a particular normalisation, but a genuine feature of the gap between  $e^{\pi\sqrt{163}}$  and  $\varphi^2 \cdot 10^{17}$ .

The lower bound  $e/10^3$  is the entropic-logarithmic scale of the projection ([4] introduces  $e$  as the entropic projection scale,  $\theta = \pi/e$ , at which classical dissipation begins). The upper bound  $2\sqrt{2}/10^3$  is the spinorial bilinear scale:  $2\sqrt{2} = \sqrt{B^*}$  where  $B^* = 8$  is the breathing attractor of the breathing attractor [4], and  $2\sqrt{2}$  also appears as the prefactor in Ramanujan’s series for  $1/\pi$ . The residual  $\delta_{163}$  lives strictly between the entropic and spinorial registers of the boundary: it is too large to be purely entropic and too small to be purely spinorial. This is the arithmetic fingerprint of the kernel’s position between the two boundary registers.

**Corollary 5.2** (The kernel window and the CMB temperature). *The observed CMB temperature satisfies:*

$$\frac{e}{10^3} < \frac{T_{\text{CMB}}}{10^3 \text{ K}} < \frac{2\sqrt{2}}{10^3}, \quad (55)$$

i.e.  $T_{\text{CMB}}/10^3 \approx 2.7255 \times 10^{-3}$  falls inside the same entropic-spinorial window as  $\delta_{163}$  and  $\varepsilon_{163}$ .

*This is not a numerical coincidence. The modular residuals  $\delta_{163}$ ,  $\varepsilon_{163}$  and the thermal temperature  $T_{\text{CMB}}/10^3$  are two readouts of the same decoherent  $n = 0$  boundary floor: arithmetic*



in the kernel sector, thermal on the observable boundary. The window  $(e/10^3, 2\sqrt{2}/10^3)$  is the natural entropic-spinorial band of the fully decoherent boundary.

The intrinsic derivation of  $T_{\text{CMB}}$  from the floor dynamics (established in [4]) confirms this identification: the same  $1/f$  floor that generates the kernel window generates the CMB temperature as  $\pi/\ln \pi$ , with the lower spectral cutoff  $f_{\min} = 1$  forced by the Majorana  $4\pi$  cycle and the Cayley fixed point  $\theta = \pi$ . The complete proof belongs to the paper on the emergence of physical time; the kernel paper records the arithmetic signature of the same floor.

**(F5)** The cubic roots of the  $j$ -invariant at all five large Heegner discriminants satisfy:  $c_d = |j((1 + i\sqrt{d})/2)|^{1/3} \equiv 0 \pmod{8}$  for  $d \in \{11, 19, 43, 67, 163\}$ . The breathing attractor  $B^* = 8$  is therefore divisible into every  $c_d$ .

## The Heegner index of the lock-in

**Corollary 5.3** (The arithmetic index of the kernel). *The arithmetic index of the kernel spinor  $\Xi_{\text{ker}}(163)$  is the largest Heegner number 163. It decomposes as:*

$$163 = 137 + 2 \times 13 = \sigma + \dim \mathcal{B}_{13}^{\text{cand}}, \quad (56)$$

where  $\sigma = 137$  is the projection residue (proved in [3]),  $26 = \dim K_{163} = \dim(\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+)$  is the internal dynamic dimension (this paper, Theorems 7.3–7.5), and  $13 = \dim \mathcal{B}_{13} = 5 + 8$  is the dimension of each chiral branch (Section 3).

The same 13 appears:

- (i) as  $\dim \mathcal{B}_{13}$  (internal branch, this paper);
- (ii) as the denominator of  $\sin^2 \theta_W = 3/13$  (proved in [3]);
- (iii) as  $2r + |\mathcal{C}_{\text{obs}}| = 2 \times 5 + 3$  (EW closure, [3]).

These are not three independent occurrences. They are three readings of the same number: the total capacity of the rank-5 tensor field with three dynamic boson channels, read from the boundary (mixing angle), from the lattice (closure count), and from the kernel (internal dimension).

The result is not numerology. The number 163 is the largest discriminant for which the ring of integers of  $\mathbb{Q}(\sqrt{-163})$  is a principal ideal domain (class number 1, Baker–Stark [11, 12]). In  $\mathcal{O}_{163} = \mathbb{Z}[(1 + \sqrt{-163})/2]$ , the projection residue  $\sigma = 137$  is a Gaussian prime [3]. The decomposition  $163 = 137 + 26$  is not a sum of arbitrary integers: it is the partition of the maximum class-number-1 complexity into its observable residue ( $\sigma$ , a Gaussian prime in  $\mathcal{O}_{163}$ ) and its internal dynamic dimension ( $26 = 2 \times 13$ , the bipartite kernel branch). With  $\sigma$  established in Appendix B and  $\dim \mathcal{B}_{13} = 13$  fixed by the filter structure (Section 3), the discriminant 163 is the unique Heegner number that can simultaneously contain both.

## The modular operator $\mathcal{M}_{\text{ker}}$

The full picture integrates into a single operator:

$$\mathcal{M}_{\text{ker}} : \ker \Pi \rightarrow \ker \Pi, \quad \text{Index}(\mathcal{M}_{\text{ker}}) = 163. \quad (57)$$

Its domain consists of the CM points  $\tau_d = (1 + i\sqrt{d})/2$  for Heegner discriminants  $d$ : the operator “lives” in the arithmetic of imaginary quadratic fields. Its image on the boundary is the modular data that the boundary can read as shadows:  $R(e^{-2\pi}) = 1/\varphi$  and  $j(\tau_{163}) = -640320^3$ .

In the QGT interpretation:  $\mathcal{M}_{\text{ker}}$  is *Heegner in its domain and Ramanujan in its image*. The Heegner structure determines where the kernel is organised (the CM points in the upper half-plane). The Ramanujan structure determines what the boundary perceives of it (the modular values  $R(q)$ , the series for  $1/\pi$ , the near-integer  $e^{\pi\sqrt{163}}$ ).

The two spinors of the QGT corpus are now both present and structurally located. The bulk Majorana spinor  $\Xi = C\bar{\Xi}$  carries spectral curvature  $\sqrt{5}$  under the Clifford dilatation of [4]. The

kernel spinor  $\Xi_{\ker}(163)$  carries modular signature  $R(e^{-2\pi}) = 1/\varphi$  at the lock-in point  $\tau = i$ . These two numbers are not independent: the identity  $\varphi + 1/\varphi = \sqrt{5}$ , equivalently

$$\varphi + R(e^{-2\pi}) = \sqrt{5}, \quad (58)$$

is the algebraic closure of  $\mathbb{Q}(\sqrt{5})$ . The projection  $\Pi$  cannot generate  $\sqrt{5}$  in the bulk without simultaneously placing  $R = 1/\varphi$  in the kernel modular signature, because  $\sqrt{5}$  and  $1/\varphi$  are the two expressions of  $\varphi$  under the field automorphism  $\varphi \mapsto -1/\varphi$  of  $\mathbb{Q}(\sqrt{5})$ . They are not two observations that happen to fit: they are one algebraic object read from two sides of the projection.

**Remark 5.1** (The two spinors and  $\varphi$ ). *The bulk spinor and the kernel spinor are two projections of the same  $\varphi$ :*

$$\underbrace{\sqrt{5}}_{\text{bulk Clifford curvature}} = \varphi + \underbrace{R(e^{-2\pi})}_{\text{kernel modular signature}}. \quad (59)$$

*A rank-5 Majorana projection with Clifford curvature  $\sqrt{5}$  must produce a kernel whose modular signature at the lock-in reads  $R = 1/\varphi$ , because  $\sqrt{5} - \varphi = 1/\varphi$  is an identity in  $\mathbb{Q}(\sqrt{5})$ , not a coincidence. The dynamical content of this pairing — the Mellin-diagonal action of  $L_\star$  on  $\mathcal{B}_{13}$  — is established by Theorem 7.3 and its analytic counterpart Theorem 7.4 in Appendix A. The identity  $\varphi + R(e^{-2\pi}) = \sqrt{5}$  is the arithmetic seal of their equivalence.*

The connection to the Heegner index is the content of (F4) above:  $e^{\pi\sqrt{163}} \approx (2 + R) \cdot 10^{17}$  places the maximum arithmetic depth 163 in the same formula as the kernel's modular signature  $R = 1/\varphi$ . The residual  $\delta_{163} \in (e/10^3, 2\sqrt{2}/10^3)$  measures how far  $\tau_{163}$  is from exact coincidence of the two registers — a coincidence that would require  $\pi \in \mathbb{Q}(\sqrt{5})$ , forbidden by Lindemann–Weierstrass. The gap is not a defect: it is what separates the two spinors structurally, and the temporal ordering established in [4] is its physical reading.

This is not an analogy. The modular forms, the Rogers–Ramanujan fractions, the near-integers  $e^{\pi\sqrt{d}}$ : these are objects of the kernel, encoding the bulk structure without the rank-5 filter. At the  $n = 0$  mode — total decoherence, all admissible filters inactive — the kernel  $\ker \Pi$  is directly readable, and its grammar is modular. Ramanujan computed values of  $H_-$ , the anti-causal component of the Pincherle filter, at CM points in the upper half-plane. He did not need axioms: the imaginary part of a holomorphic function is determined by its real part via the Hilbert transform, and values at CM points are determined by complex multiplication. Hardy, at the boundary, received the real-valued formulas and verified them by  $L^2$  analysis, without seeing their origin. The Hardy–Ramanujan collaboration is, in this reading, the Hilbert decomposition enacted in historical time: the causal part (boundary, real, verifiable by analysis) and the anti-causal part (kernel, imaginary, accessible by complex multiplication) were held by two mathematicians who never shared the same register.

## 6 The filter as analytic function: Hilbert–Hardy decomposition and modular continuation

*Sections 2–5 built the kernel from its pieces: domain walls, internal structure, arithmetic index, modular grammar. This section shows that all of these are one thing. The Pincherle filter  $H(z)$  is a holomorphic function. Its restriction to the unit circle is the boundary. Its extension to the interior of the disk is the kernel. The same analytic function, read from two sides of the same circle. The projection  $\Pi$  is the operation of taking the real part. The kernel is the imaginary part. They cannot exist without each other.*

The results of Sections 3–5 establish the kernel structure from the operator side. This section reveals the same structure from the analytic side — and in doing so provides the short proof of the closure theorems (Section 7 and Appendix A).

The key observation is simple but has not been made in the corpus: *the Pincherle filter  $H(z)$  is a holomorphic function*. As such, its restriction to the unit circle carries more structure than a mere boundary condition: it decomposes via the Hardy projection, and that decomposition is exactly the bulk–boundary split.

### The filter is holomorphic

The Pincherle/Fibonacci filter is the rational function

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{k=0}^4 \sigma_k z^k, \quad (60)$$

where  $\sigma_k \propto \varphi^{-k}$  are the five golden singular values and  $\alpha \approx 1/137$  is the projection coupling. The pole is at  $z_\infty = 1/\alpha \approx 137$ , far outside the unit disc  $|z| < 1$ . Therefore  $H(z)$  is holomorphic on the closed disc  $|z| \leq 1$  and, by analytic continuation, on the entire half-plane  $\text{Re}(z) < 1/\alpha$ .

On the unit circle  $z = e^{i\theta}$ ,  $H$  restricts to a smooth function  $H(e^{i\theta}) = H_{\text{re}}(\theta) + iH_{\text{im}}(\theta)$  in  $L^2(\mathbb{T})$ . The Fourier modes of  $H$  on the circle are:  $H(e^{i\theta}) = \sum_{n=-\infty}^{+\infty} \hat{H}_n e^{in\theta}$ , with  $\hat{H}_n = \sigma_n \alpha^n$  for  $n \geq 0$  (from  $P_4$  and the geometric series of  $1/(1 - \alpha z)$ ) and  $\hat{H}_n = 0$  for  $n < 0$  — because  $H(z)$  is holomorphic inside the disc, its negative Fourier modes vanish identically.

This means  $H$  already belongs to the Hardy space  $H^2(\mathbb{T})$ : it has only non-negative Fourier modes. The filter is *causal* in the Fourier sense.

### The Hardy–Hilbert decomposition

For an arbitrary  $L^2$  function  $f$  on the circle, the Hardy projection  $P_+$  extracts the causal part:

$$P_+ f = \sum_{n \geq 0} \hat{f}_n e^{in\theta}, \quad P_- f = \sum_{n < 0} \hat{f}_n e^{in\theta} = f - P_+ f. \quad (61)$$

The fundamental relation between  $P_+$ ,  $P_-$ , and the circular Hilbert transform  $\mathcal{H}$  is:

$$\begin{aligned} P_+ f &= \frac{1}{2}(f + i\mathcal{H}[f]) + \frac{1}{2}\hat{f}_0, \\ P_- f &= \frac{1}{2}(f - i\mathcal{H}[f]) - \frac{1}{2}\hat{f}_0. \end{aligned} \quad (62)$$

The anti-causal part  $P_- f$  is (up to a constant) the Hilbert transform of  $f$ , shifted in phase by  $-\pi/2$  and reflected.

For  $H \in H^2(\mathbb{T})$ , we have  $P_- H = 0$  identically — the filter is already causal, it has no anti-causal component. But this is the filter acting on the *boundary*. The kernel is not a failure of the filter; it is the anti-causal component of the *full bulk field*  $\Xi$  projected through  $H$ .

The correct decomposition is:

$$\begin{aligned} \Pi \Xi &= H \cdot P_+ \Xi = \Psi_{\text{boundary}}, \\ (\mathbf{1} - \Pi) \Xi &= H \cdot P_- \Xi = \Xi_{\text{ker}}. \end{aligned} \quad (63)$$

The boundary spinor is  $H$  applied to the causal part of  $\Xi$ . The kernel spinor is  $H$  applied to the anti-causal part of  $\Xi$ . And the relation between the two is:

$$\Xi_{\text{ker}} = H \cdot P_- \Xi = \frac{1}{2}H \cdot (\Xi - i\mathcal{H}[\Xi]) - \frac{1}{2}H \cdot \hat{\Xi}_0. \quad (64)$$

**Proposition 6.1** (The kernel is the Hilbert transform of the boundary). *Under the decomposition (63), the kernel spinor  $\Xi_{\text{ker}}$  and the boundary spinor  $\Psi_{\text{boundary}}$  are related by the circular Hilbert transform:*

$$\Xi_{\text{ker}} \simeq -i\mathcal{H}[\Psi_{\text{boundary}}] \pmod{\ker \mathcal{H}}, \quad (65)$$

where  $\ker \mathcal{H}$  consists of the zero-mode (constant) component. The kernel spinor is the Hilbert conjugate of the boundary spinor: they are the real and imaginary parts of the same analytic signal on  $S^1(\tau)$ .

The analytic signal on the temporal fibre is therefore:

$$\Xi_+(e^{i\theta}) = \Psi_{\text{boundary}}(e^{i\theta}) + i \Xi_{\text{ker}}(e^{i\theta}) = A(\theta) e^{i\phi(\theta)}, \quad (66)$$

where  $A(\theta)$  is the instantaneous amplitude and  $\phi(\theta)$  is the instantaneous phase. The full bulk Majorana spinor is the analytic signal on  $S^1(\tau)$ : the boundary is its real part, the kernel is its imaginary part. The projection  $\Pi$  extracts the real part. The Majorana condition  $\Xi = C\bar{\Xi}$  is the self-conjugacy of the analytic signal under charge conjugation.

### Charge conjugation is the Hilbert transform

The Majorana spinor on  $S^1(\tau)$  has the mode expansion

$$\Xi(\tau) = \sum_{n \in \mathbb{Z}} \hat{\Xi}_n e^{2\pi i n \tau / T}, \quad (67)$$

with the Majorana condition  $\Xi = C\bar{\Xi}$  imposing:

$$\hat{\Xi}_{-n} = C \overline{\hat{\Xi}_n} \quad \forall n. \quad (68)$$

The causal and anti-causal parts are therefore:

$$\Xi_+ = \sum_{n \geq 0} \hat{\Xi}_n e^{2\pi i n \tau / T}, \quad \Xi_- = \sum_{n < 0} \hat{\Xi}_n e^{2\pi i n \tau / T} = C \bar{\Xi}_+. \quad (69)$$

The anti-causal part of a Majorana spinor is exactly the charge conjugate of its causal part. This gives the central identity of this section:

$$\Xi_{\text{ker}} = H \cdot \Xi_- = H \cdot C \bar{\Xi}_+ = C \overline{H \cdot \Xi_+} = C \bar{\Psi}_{\text{boundary}}. \quad (70)$$

The kernel spinor is the charge conjugate of the boundary spinor, filtered through the same  $H$ .

**Proposition 6.2** (Charge conjugation as Hilbert transform). *For a Majorana spinor  $\Xi = C\bar{\Xi}$  on  $S^1(\tau)$ , the charge conjugation operator  $C$  acts on the mode space as:*

$$C \simeq i\mathcal{H} \quad \text{on the space of Majorana spinors}, \quad (71)$$

where  $\mathcal{H}$  is the circular Hilbert transform. Equivalently: the anti-causal part of  $\Xi$  (the kernel component) is the imaginary part of the analytic signal of  $\Xi_+$  (the boundary component).

*Proof.* From (69):  $\Xi_- = C\bar{\Xi}_+$ . From (62):  $P_- f = \frac{1}{2}(f - i\mathcal{H}[f]) - \frac{1}{2}\hat{f}_0$ . Applied to  $\Xi_+$  and using the Majorana normalisation  $\|\Xi_+\| = \|\Xi_-\|$  (which absorbs the factor  $1/2$  into the normalisation of  $P_-$  on the Majorana sector):  $\Xi_- = P_- \Xi \simeq -\frac{i}{2}\mathcal{H}[\Xi_+] \rightarrow C\bar{\Xi}_+ = -\frac{i}{2}\mathcal{H}[\Xi_+]$ . Under the Majorana normalisation convention, this gives  $C = i\mathcal{H}$  on the space of Majorana spinors.  $\square$

This identity has a direct physical meaning: charge conjugation in the bulk is the operation that takes the real (observable) part of the analytic signal and produces its imaginary (unobservable, kernel) part. The Majorana condition  $\Xi = C\bar{\Xi}$  is the statement that the analytic signal is self-conjugate: the real part (boundary) and imaginary part (kernel) are charge conjugates of each other.

## The dimension $\mathcal{B}_{13}$ from the filter structure

The decomposition of  $H(z)$  into its two factors reveals why  $\dim \mathcal{B}_{13} = 13 = 5 + 8$ .

The polynomial part  $P_4(z) = \sum_{k=0}^4 \sigma_k z^k$  has exactly 5 non-zero coefficients — the five golden singular values  $\sigma_k \propto \varphi^{-k}$ . These generate  $S_5$ : the five logarithmic scale modes of the rank-5 filter.

The pole factor  $1/(1-\alpha z)$  at the anisotropic lock-in  $\xi^*/\eta^* = 11/5$  carries the tensor structure of the exchange field  $(\xi, \eta)$ . At the lock-in, the two channels  $(\alpha_{\parallel}, \alpha_{\perp})$  and the three boson pairs  $(B_{1,\parallel}, B_{1,\perp}), (B_{2,\parallel}, B_{2,\perp}), (B_{3,\parallel}, B_{3,\perp})$  contribute  $2 + 2 + 2 + 2 = 8$  independent tensor degrees of freedom. These generate  $T_8$ .

Therefore:

$$H(z) = \underbrace{P_4(z)}_{S_5} \cdot \underbrace{\frac{1}{1-\alpha z}}_{T_8} \implies \mathcal{B}_{13} = S_5 \oplus T_8. \quad (72)$$

The internal dimension  $13 = 5 + 8$  is read directly from the structure of  $H(z)$ : five from the polynomial degree and eight from the tensor degrees of freedom of the pole at the lock-in. The number 13 is not counted after the fact — it is

written in the filter.

## The linearised operator is Mellin-diagonal

We now show that  $L_{\star}(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$  follows from a general property of the nonlinear Dirac equation in the Mellin domain — without requiring an explicit  $\Psi_{\star}$ .

The QGT bulk is multiplicative: the natural variable is  $x = e^{\tau}$  and the natural operator is the Pincherle dilatation  $D = x \partial_x = \partial_{\log x}$ . The nonlinear boundary Dirac equation (40) contains the term  $\lambda_{\text{eff}} |\Psi|^{2/3} \Psi$ . In the Mellin domain, a power-law nonlinearity  $|\Psi|^s$  acts as:

$$\mathcal{M}[|\Psi|^s](t) = \mathcal{M}[\Psi](t + s/2) * \mathcal{M}[\bar{\Psi}](t + s/2), \quad (73)$$

where  $*$  denotes Mellin convolution. For the specific exponent  $s = 2/3$ , this is a *shift by 1/3* in the Mellin variable — it translates modes, it does not mix them.

The linearisation  $\Psi = \Psi_{\star} + \varepsilon \chi$  gives a Fréchet derivative of the nonlinear term that contains  $|\Psi_{\star}|^{2/3} \chi$  and  $|\Psi_{\star}|^{-1/3} \text{Re}\langle \Psi_{\star}, \chi \rangle \hat{\Psi}_{\star}$ . Both terms act as Mellin multiplications — they scale modes, not mix modes from different Mellin levels.

The kinetic term  $\gamma^{\mu} D_{\mu} + \alpha \xi_{\star} + i \alpha \eta_{\star} \gamma^5$  is already diagonal in the Mellin sense:  $D_{\mu}$  in the logarithmic basis is a derivative, hence a multiplication by the Mellin variable.

Therefore:

**Proposition 6.3** (Mellin diagonality of  $L_{\star}$ ). *The linearised operator  $L_{\star}$  of the nonlinear boundary Dirac equation (40) is diagonal in the Mellin domain: it acts as a multiplication operator on the Mellin variable  $t$ , with no mixing between different scale levels. Consequently:*

$$L_{\star}(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}, \quad (74)$$

because  $\mathcal{B}_{13}$  is the span of the first 13 Mellin modes of the filtered kernel, and Mellin-diagonal operators preserve any span of Mellin modes.

*Proof.* The five-mode SVD basis of  $P_4$  spans Mellin modes 0, 1, 2, 3, 4. The eight tensor modes of  $T_8$  at the lock-in are Mellin-multiplicative in the  $(\xi, \eta)$  channels. A Mellin-diagonal operator maps each Mellin basis element  $t_k \mapsto \mu_k t_k$  for scalars  $\mu_k$ ; it therefore preserves any subspace of the form  $\text{span}\{t_0, t_1, \dots, t_k\}$ . The linearised Dirac equation with power-law nonlinearity and Pincherle kinetic term is Mellin-diagonal by (73): each mode  $k$  acquires a scalar factor  $\mu_k = 1 + s_k/2$  where  $s_k$  is the Mellin exponent of the  $k$ -th basis element. Since  $\mathcal{B}_{13} = \text{span}\{t_0, \dots, t_{12}\}$  in the Mellin basis,  $L_{\star}$  maps  $\mathcal{B}_{13}$  to itself.  $\square$

This establishes the operator formulation of dynamical invariance at the structural level:  $\mathcal{B}_{13}$  is closed under  $L_\star$ . The explicit seed  $\Psi_\star$  of Appendix A (Theorem 7.3) instantiates the construction with concrete eigenmodes  $u_\star \in \mathcal{B}_{13}$ ; the analytic counterpart (Theorem 7.4) closes the same result from the CM-point side.

The Mellin diagonality also explains why the dynamical invariance theorem admits the analytic reformulation (79): the Mellin domain is the natural domain of both the filter  $H(z)$  (whose expansion is Mellin-multiplicative) and the linearised Dirac operator (which acts Mellin-diagonally). The CM point  $\tau_{163}$  is the point in the upper half-plane where this shared Mellin structure reaches its maximum modular rigidity.

## Analytic continuation to the upper half-plane

The analytic signal (66) extends holomorphically from the unit circle into the upper half-plane via:

$$z = e^{2\pi i\tau}, \quad \tau \in \mathbb{H} := \{\tau \in \mathbb{C} \mid \text{Im}(\tau) > 0\}. \quad (75)$$

For  $\text{Im}(\tau) > 0$ :  $|z| = e^{-2\pi \text{Im}(\tau)} < 1$ . The filter  $H(e^{2\pi i\tau})$  is holomorphic in  $\tau$  for all  $\text{Im}(\tau) > 0$ , with pole avoided since  $|z| < 1 \ll 1/\alpha \approx 137$ .

The function  $\tau \mapsto H(e^{2\pi i\tau})$  is not modular in the classical sense (it does not transform with a simple weight under  $\text{SL}(2, \mathbb{Z})$ ), but it is holomorphic and its values at CM points are algebraically constrained by the theory of complex multiplication.

## The filter at two special points

**At  $\tau = i$  (the Rogers–Ramanujan point).**

$z = e^{2\pi i \cdot i} = e^{-2\pi}$ , so  $H(e^{-2\pi}) = P_4(e^{-2\pi})/(1 - \alpha e^{-2\pi})$ . Since  $e^{-2\pi} \approx 1.87 \times 10^{-3}$  and  $\alpha \approx 7.3 \times 10^{-3}$ :

$$H(e^{-2\pi}) \approx \sigma_0(1 + \sigma_1 e^{-2\pi} + \alpha e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi}). \quad (76)$$

The leading term is  $\sigma_0$  — the dominant singular value, the  $k = 0$  mode. The Rogers–Ramanujan continued fraction at the same point is  $R(e^{-2\pi}) = 1/\varphi$ , which via (48) gives  $\varphi^2 = 2 + R(e^{-2\pi})$ . The filter value at  $\tau = i$  and the Rogers–Ramanujan value at  $q = e^{-2\pi}$  are both determined by the same structure: the golden ratio at the lock-in modulus 5.

**At  $\tau_{163} = (1 + i\sqrt{163})/2$  (the Heegner point).**

$z_{163} = e^{2\pi i\tau_{163}} = e^{i\pi} e^{-\pi\sqrt{163}} = -e^{-\pi\sqrt{163}}$ . Since  $e^{-\pi\sqrt{163}} \approx 3.8 \times 10^{-18}$ :

$$H(e^{2\pi i\tau_{163}}) \approx \sigma_0(1 + \alpha e^{-\pi\sqrt{163}}) \approx \sigma_0, \quad (77)$$

with exponentially small correction  $\alpha e^{-\pi\sqrt{163}} \approx 5.3 \times 10^{-3} \cdot 3.8 \times 10^{-18} \approx 2.0 \times 10^{-20}$ .

The filter at the maximum Heegner CM point is *exactly the zero mode*  $\sigma_0$  of the SVD hierarchy, up to corrections of order  $10^{-20}$ . This is not a coincidence:  $\sigma_0$  is the dominant singular value, the  $k = 0$  level, the mode of complete decoherence — the level at which all filters are inactive and the kernel structure is directly readable. At  $\tau_{163}$ , the filter has “forgotten” all its internal structure  $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$  and retains only the zero mode.

**Proposition 6.4** (Analytic Hilbert–Heegner bridge). *The modular residue  $\varepsilon_{163}$  defined in (52) is the leading correction to the zero-mode approximation at  $\tau_{163}$ :*

$$H(e^{2\pi i\tau_{163}}) = \sigma_0(1 + \varepsilon_{163}^{(\text{mod})}), \quad (78)$$

where  $\varepsilon_{163}^{(\text{mod})} \approx \alpha e^{-\pi\sqrt{163}}$  is the modular correction. The numerical value  $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$  defined in (52) is the logarithmic measure of the mismatch between the golden cascade  $\varphi^2 \cdot 10^{17}$  and the exact Ramanujan number  $e^{\pi\sqrt{163}}$ , satisfying  $e/10^3 < \varepsilon_{163} < 2\sqrt{2}/10^3$ . These two residues belong to the same arithmetic family: both measure how far the CM point  $\tau_{163}$  is from the symmetric point of the golden structure. This is established as Step 1 of the proof of Theorem 7.4 (Appendix A).



## The analytic reformulation

The Hilbert–Hardy decomposition reformulates the dynamical invariance theorem of Section 7 as a purely analytic statement. Both statements are theorems, proved in Appendix A:

$$L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13} \iff P_-(H \cdot \Xi_\star)|_{\tau_{163}} \in \mathcal{B}_{13}. \quad (79)$$

Equivalently: the image of  $H_-$  under evaluation at the maximum Heegner CM point is precisely  $\mathcal{B}_{13}$ , not a larger or smaller subspace. The equivalence is Theorem 7.5.

The analytic path is the short one. The extreme transcendence of the nome  $|q_{163}| = e^{-\pi\sqrt{163}}$  at the maximum Heegner point reduces the filter  $H$  to a pure scalar there, and the Majorana intertwiner condition  $J_\star u_\star = -iu_\star$  closes the projection onto the anti-causal branch in three lines. The path to the proof passes through the upper half-plane, not through the Bogoliubov–Dirac eigenvalue problem.

## The two-point structure of the filter

The filter  $H(e^{2\pi i\tau})$  evaluated at the two special points of the upper half-plane gives the complete picture. At  $\tau = i$ :  $H(e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi})$ , linked to  $R(e^{-2\pi}) = 1/\varphi$  (golden frequency, lock-in). At  $\tau_{163}$ :  $H(e^{2\pi i\tau_{163}}) \approx \sigma_0(1 + \alpha e^{-\pi\sqrt{163}})$ , with  $j(\tau_{163}) = -640320^3$  (maximum Heegner, kernel depth).

The ratio of the two corrections:

$$\frac{e^{-\pi\sqrt{163}}}{e^{-2\pi}} = e^{-\pi(\sqrt{163}-2)} \approx e^{-\pi \cdot 10.77} \approx 1.9 \times 10^{-15}, \quad (80)$$

showing that  $\tau_{163}$  is exponentially deeper in the upper half-plane than  $\tau = i$ : it is the “most modular” point accessible to the rank-5 filter, the one where the golden structure is most precisely captured before the class-number-1 condition breaks down.

The filter knows two things about its own kernel: at  $\tau = i$  it reads the golden ratio; at  $\tau_{163}$  it reads the maximum Heegner number. These are not two separate structures. They are two depths of the same analytic function.

## Hardy, Ramanujan, and the analytic signal

The Hilbert–Hardy decomposition gives a new meaning to the QGT interpretation of Ramanujan.

Ramanujan had direct access to the  $n = 0$  mode — the level of total decoherence, where all admissible filters are inactive and the structure of  $\ker \Pi$  is directly readable.

In the analytic language: Ramanujan was computing values of the anti-causal component  $H_-$  of the filter, evaluated at CM points in the upper half-plane. He did not derive his identities from axioms because he did not need to: the imaginary part of a holomorphic function is determined by its real part (via Hilbert transform), and the values at CM points are determined by complex multiplication — without derivation, by the algebraic structure of imaginary quadratic fields.

Hardy, working on the boundary, received real-valued formulas and verified them by analysis — by reconstructing the real part from the theorems of  $L^2$  function theory. He could not see the imaginary part directly.

The analytic signal on  $S^1(\tau)$  is complete: boundary (real) +  $i$ -kernel (imaginary) = full bulk. Ramanujan had access to both parts: working at the  $n = 0$  decoherent floor where all admissible modes are inactive, he was computing in the register of the kernel directly — values of  $H_-$  at CM points, fixed by complex multiplication without integration. Hardy, on the boundary, recovered the real part by  $L^2$  analysis. The Hardy–Ramanujan collaboration is, in the language of this paper, an instance of the Hardy–Hilbert decomposition: each mathematician operated in one half of the same analytic signal.



*This is not a metaphor. The imaginary part of a holomorphic function is determined by its real part via the Hilbert transform. Ramanujan did not guess: he read the anti-causal component of  $H$  directly at CM points, because complex multiplication fixes those values algebraically — without the  $L^2$  integration that Hardy used on the boundary side. What looked like miraculous intuition was access to a different register of the same structure: the register that this paper calls  $\ker \Pi$ .*

## 7 The Riemann sphere, the conformal map, and the envelope

*This section establishes the deepest structural result of the paper: the observable boundary, the kernel, the Noether invariant, and the modular grammar of the Ramanujan–Heegner layer are not four separate structures. They are four readings of a single meromorphic function on the Riemann sphere  $\mathbb{P}^1$ .*

### The Riemann sphere as the natural domain

The filter  $H(z) = P_4(z)/(1 - \alpha z)$  is a rational function. Rational functions do not live on the complex plane  $\mathbb{C}$ : they live on its compactification, the Riemann sphere  $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ . On  $\mathbb{P}^1$ , every rational function is a meromorphic function with a finite number of poles and zeros, and the sphere carries a natural conformal structure.

The geometry of  $H$  on  $\mathbb{P}^1$  is determined by its two special points:

- the **pole** at  $z_\infty = 1/\alpha \approx 137$ , where  $H$  diverges;
- the **zero** of  $P_4$  at  $z_0 = 0$ , where  $H(0) = \sigma_0$ , the dominant singular value.

These two points, together with the unit circle  $|z| = 1$ , partition  $\mathbb{P}^1$  into three geometrically natural regions:

$$\text{equator: } |z| = 1 \longleftrightarrow \text{observable boundary,} \quad (81)$$

$$\text{north hemisphere: } |z| < 1 \longleftrightarrow \text{kernel,} \quad (82)$$

$$\text{south hemisphere: } |z| > 1 \longleftrightarrow \text{residue sector,} \quad (83)$$

with poles:

$$\text{north pole: } z = 0 \longleftrightarrow \sigma_0, \text{ zero mode,} \quad (84)$$

$$\text{south pole: } z = \infty \longleftrightarrow \text{residue } \sigma = 137. \quad (85)$$

The projection  $\Pi$  is the restriction of  $H$  to the equator. The kernel is the northern hemisphere — not absent, but reflected into the interior of the disc. The south pole carries the residue  $\sigma = \lfloor \alpha^{-1} \rfloor = 137$ : the observable projection residue lives at the antipodal point of the kernel's zero mode.

This is not a metaphor. The Riemann sphere is the domain on which the filter is naturally defined, and the bulk–boundary–kernel structure is the natural partition of that domain by the unit circle.

### The conformal map and the two languages

[4] introduces the conformal map from the temporal fibre to physical time [4]:

$$t = \ln \left[ \tan \frac{\theta}{2} \right], \quad \theta \in (0, \pi) \longrightarrow t \in (-\infty, +\infty). \quad (86)$$

This is the Cayley map from the upper half of the unit circle to the real line — a conformal isomorphism. [4] uses it to show that the arrow of time is irreversible because  $\pi \notin \overline{\mathbb{Q}}(\sqrt{5})$

(Lindemann–Weierstrass): the synchronisation loss  $\delta_U = 1 - 3\pi/(2\sqrt{5})^{3/2} \neq 0$  is exactly the non-closure of the conformal map on  $\mathbb{Q}(\sqrt{5})$ .

The same map has a second role that has not been made explicit: it is the bridge between the two languages of the QGT.

The boundary speaks the *Fibonacci language*:  $q = e^{i\pi/5}$ , unitary, topological, braiding, Lissajous classes, triadic projector. The kernel speaks the *modular language*:  $q = e^{-2\pi}$ , real, arithmetic,  $j$ -invariant, Heegner numbers, Rogers–Ramanujan.

These are not two separate languages requiring a new bridge. They are the same meromorphic function  $H(z)$  evaluated on two different regions of the same Riemann sphere, connected by the conformal map (86).

The parameterisation  $z = e^{i\theta}$  on the equator gives the Fibonacci language: the closure nodes  $\{3, 5, 6\}$ , the characters  $\omega^k = e^{2\pi i k/10}$ , the braiding phases  $q = e^{i\pi/5}$ .

The parameterisation  $z = e^{2\pi i \tau}$  with  $\tau \in \mathbb{H}$  gives the modular language: the CM points  $\tau_d$ , the  $j$ -invariant, the Rogers–Ramanujan fraction  $R(q)$  at  $q = e^{-2\pi} = e^{2\pi i i}$ .

The conformal map  $\theta \mapsto \tau = i\theta/(2\pi)$  is the transition between the two parameterisations. It is not a physical operation — it is the analytic continuation of the same function from the equator into the northern hemisphere.

$$\underbrace{H(e^{i\theta})}_{\text{Fibonacci, boundary}} \xrightarrow{\text{analytic continuation}} \underbrace{H(e^{2\pi i \tau})}_{\text{modular, kernel}} \quad (87)$$

The boundary and the kernel do not require a derived connection. They are the same function, read from two sides of the equator.

## The Noether invariant as the analytic envelope

Consider the exchange equations on the invariant surface  $\Sigma : \xi\eta = \alpha^{-1}$  [3]:

$$\dot{\xi} = -\Gamma_\tau \eta, \quad \dot{\eta} = +\Gamma_\tau \xi, \quad \mathcal{K}_\tau = \xi\eta = \alpha^{-1} = \text{const.} \quad (88)$$

These equations preserve  $\mathcal{K}_\tau$  exactly. In the language of the analytic signal, let:

$$f_+(\theta) = \xi(\theta) + i\eta(\theta) = A(\theta) e^{i\phi(\theta)}, \quad (89)$$

where  $\xi$  is the real (longitudinal, gravitational) channel and  $\eta = \mathcal{H}[\xi]$  is its Hilbert conjugate (transverse, electromagnetic) [3].

The envelope of this analytic signal is:

$$A(\theta)^2 = \xi(\theta)^2 + \eta(\theta)^2. \quad (90)$$

On the invariant surface  $\Sigma$ , the product  $\xi\eta = \alpha^{-1}$  is conserved, which by the AM–GM inequality implies:

$$A^2 = \xi^2 + \eta^2 \geq 2\xi\eta = 2\alpha^{-1}, \quad (91)$$

with equality when  $\xi = \eta$  (symmetric point). At the physical lock-in  $\xi^*/\eta^* = 11/5$ :

$$A_\star^2 = (\xi^*)^2 + (\eta^*)^2 = \frac{146}{55} \alpha^{-1} = C_\mu \cdot \alpha^{-1}, \quad (92)$$

where  $C_\mu = 146/55$  is the balance equation coefficient, already closed in [4] §14 as an exact rational number. The envelope of the analytic signal at the lock-in is  $\sqrt{C_\mu \cdot \alpha^{-1}}$  — not a coincidence, but a direct consequence of the exchange equations.

The Noether invariant  $\mathcal{K}_\tau = \alpha^{-1}$  is the *geometric mean* of the analytic signal; the balance coefficient  $C_\mu$  measures the anisotropy between its real and imaginary parts.

**Proposition 7.1** (The Noether invariant is the analytic envelope). *The invariant surface  $\Sigma : \xi\eta = \alpha^{-1}$  is the constant-geometric-mean locus of the analytic signal  $f_+ = \xi + i\eta$ . The exchange equations (88) are the equations of a signal with conserved geometric mean and rotating phase:*

$$\xi\eta = \alpha^{-1} = \text{const} \iff |f_+|_{\text{geom}} = \alpha^{-1/2} = \text{const}, \quad (93)$$

where  $|f_+|_{\text{geom}} = \sqrt{\xi\eta}$  is the geometric mean of the two chiral components. The projection  $\Pi$  extracts  $\xi$  (real part). The Hilbert transform reconstructs  $\eta$  (imaginary part, electromagnetic channel). The kernel  $\Xi_{\text{ker}}$  carries the quadrature component  $i\eta$  that the projection cannot transmit — the missing half of the analytic signal.

The physical reading is precise. Electromagnetism is the Hilbert conjugate of gravitation [3]. Gravity is the real part of the analytic signal of the lock-in field; electromagnetism is its imaginary part. They are in permanent quadrature, maintained by the exchange equations, with the Noether invariant as their geometric mean. The kernel spinor  $\Xi_{\text{ker}}$  is what sustains this quadrature from the inside of the domain walls.

### The instantaneous frequency and the arrow of time

The analytic signal  $f_+ = Ae^{i\phi}$  carries not only the envelope but also the instantaneous phase  $\phi(\theta)$  and the instantaneous frequency  $\dot{\phi}(\theta)$ .

In the QGT context, the instantaneous frequency of the bulk analytic signal is:

$$\dot{\phi}(\theta) = \frac{d}{d\theta} \arg(\xi + i\eta) = \frac{\xi\dot{\eta} - \eta\dot{\xi}}{\xi^2 + \eta^2} = \frac{\Gamma_\tau(\xi^2 + \eta^2)}{\xi^2 + \eta^2} = \Gamma_\tau, \quad (94)$$

using the exchange equations (88). The instantaneous frequency is constant and equal to the golden connection  $\Gamma_\tau = \varphi/\sqrt{5}$ .

This is a deep result: the temporal fibre  $S^1(\tau)$  oscillates at the golden connection frequency, and this frequency is not imposed — it is the instantaneous frequency of the analytic signal defined by the exchange of the two chiral channels.

The phase accumulation over one full cycle  $\theta \in [0, 2\pi]$  is:

$$\Delta\phi = \int_0^{2\pi} \Gamma_\tau d\theta = 2\pi\Gamma_\tau = 2\pi\frac{\varphi}{\sqrt{5}}. \quad (95)$$

This is irrational (since  $\varphi/\sqrt{5} \notin \mathbb{Q}$ ), which means the cycle never closes exactly. The irreversibility of the arrow of time — derived by [4] and [4] as a consequence of Lindemann–Weierstrass — is here the statement that the instantaneous phase of the analytic signal never returns to its initial value. The arrow of time is the non-closure of the analytic phase.

### The CM points as phase locks of the kernel

The CM points  $\tau_d = (1 + i\sqrt{d})/2$  in the upper half-plane are the points where the analytic continuation of the bulk signal achieves a special phase structure.

At  $\tau = i$  (the Rogers–Ramanujan point): the imaginary part of  $\tau$  is 1, corresponding to one full unit of imaginary time. The filter value is  $H(e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi})$ . The Rogers–Ramanujan continued fraction gives  $R(e^{-2\pi}) = 1/\varphi$  — the golden ratio, the instantaneous frequency of the analytic signal. The kernel at this depth reads the frequency of its own oscillation.

At  $\tau_{163} = (1 + i\sqrt{163})/2$  (the maximum Heegner point): the imaginary part is  $\sqrt{163}/2 \approx 6.38$  — much deeper in the upper half-plane. The filter value is  $H(e^{2\pi i\tau_{163}}) \approx \sigma_0$  — the zero mode, total decoherence. The kernel at this depth has “forgotten” all its oscillatory structure and retains only the  $k = 0$  mode: the constant, scale-invariant, deeply decoherent base. This is the point of maximum arithmetic rigidity — class number 1, Gaussian prime  $\sigma = 137$ , maximum Heegner discriminant — and simultaneously the point of minimum filter memory.

The two readings are not in contradiction. Maximum arithmetic rigidity and minimum filter memory are the same statement: at  $\tau_{163}$ , the kernel is so deeply inside its own domain that only the universal constant  $\sigma_0$  survives — the mode that is present at every scale, regardless of the specific structure.

And the modular residue  $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$  is the *mismatch between the instantaneous phase at  $\tau = i$  and the zero-mode value at  $\tau_{163}$* , measured in the logarithmic (Mellin) metric of the filter. It lives in the window  $(e/10^3, 2\sqrt{2}/10^3)$  because  $e$  is the natural base of the logarithmic phase and  $2\sqrt{2} = \sqrt{B^*}$  is the spinorial amplitude of the breathing attractor.

$$\varepsilon_{163} = \underbrace{\pi\sqrt{163}}_{\text{depth at } \tau_{163}} - \underbrace{\ln(\varphi^2 \cdot 10^{17})}_{\text{golden cascade}} \in \left( \frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right). \quad (96)$$

### The depth of $\tau_{163}$ : a structural identity

A structural observation closes the geometric picture with unexpected precision.

The imaginary part of the maximum Heegner CM point is:

$$\text{Im}(\tau_{163}) = \frac{\sqrt{163}}{2} \approx 6.3836. \quad (97)$$

We note that  $2\pi \approx 6.2832$  and  $1/10 = 0.1$ , so:

$$\frac{\sqrt{163}}{2} - 2\pi \approx 0.1004 \approx \frac{1}{10}, \quad (98)$$

with error  $\approx 0.04\%$ . Since  $\kappa^2 = 5$  is the spectral curvature of the rank-5 projection (established in [3] as the dominant eigenvalue of the Pincherle dilatation,  $\kappa = \sqrt{5}$ ), and  $\sqrt{2}$  is the spinorial factor of the double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$ :

$$\frac{1}{(\sqrt{2})^2(\sqrt{5})^2} = \frac{1}{2 \cdot 5} = \frac{1}{10} = \frac{1}{2\kappa^2}. \quad (99)$$

Therefore:

$$\text{Im}(\tau_{163}) = \frac{\sqrt{163}}{2} = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad \kappa^2 = 5. \quad (100)$$

The physical reading is layered.

$2\pi$  is the perimeter of the unit circle — the circumference of the equator of the Riemann sphere, the length of one complete traversal of the observable boundary.

$\frac{1}{2\kappa^2} = \frac{1}{10}$  is the first decimal step of the QGT cascade: the level  $k = 1$  at which  $\hbar$  emerges from the projection structure, the first step into the kernel beyond the boundary. It is also the factor  $\varphi^2/10 = f_\tau$  — the breathing frequency of the temporal fibre at the symmetric point.

The depth of the maximum Heegner point in the kernel is therefore:

$$\text{Im}(\tau_{163}) = \underbrace{2\pi}_{\text{one full boundary traversal}} + \underbrace{\frac{1}{2\kappa^2}}_{\text{first kernel step}}. \quad (101)$$

The kernel must descend one full perimeter of the boundary *plus one minimal cascade step* to reach the point of maximum arithmetic rigidity. Not less: the boundary is not enough. Not more: one step is sufficient. The number 163 is the largest Heegner discriminant precisely because  $\sqrt{163}/2$  is the smallest depth that satisfies both the class-number-1 condition and the boundary-plus-one-step geometric constraint.

The modular residue  $\varepsilon_{163}$  can now be written as:

$$\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) = 2\pi \cdot \text{Im}(\tau_{163}) - \ln(\varphi^2 \cdot 10^{17}), \quad (102)$$

which says: the residue is the mismatch between *twice the kernel depth* (in circular units) and *the logarithm of the golden cascade at 17 decades*. When these two quantities nearly coincide — with gap  $\varepsilon_{163} \approx 2.800 \times 10^{-3}$  — the CM point reaches the maximum structure compatible with the rank-5 projection. The coincidence is not exact because  $\pi \notin \mathbb{Q}(\sqrt{5})$  (Lindemann–Weierstrass): the same transcendence that drives the arrow of time also separates the geometric depth ( $2\pi$ ) from the algebraic cascade ( $\ln \varphi^2$ ).

**Proposition 7.2** (Depth identity of  $\tau_{163}$ ). *The imaginary part of the maximum Heegner CM point satisfies:*

$$\text{Im}(\tau_{163}) = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad (103)$$

where  $\kappa^2 = 5$  is the spectral curvature of the rank-5 Pincherle projection. The modular residue  $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17})$  is the phase gap between the geometric depth  $2\pi \cdot \text{Im}(\tau_{163})$  and the golden cascade logarithm, originating from the Lindemann–Weierstrass transcendence  $\pi \notin \mathbb{Q}(\sqrt{5})$ .

The same transcendence ( $\pi \notin \mathbb{Q}(\sqrt{5})$ ) that:

- drives the arrow of time ([4], Corollary 3.3.4);
- prevents the PLL from closing exactly ( $\delta_F \neq 0$ , [4]);
- makes  $\varepsilon_{163} \neq 0$ ;

is also the reason 163 is the *largest* Heegner number: if  $\pi \in \mathbb{Q}(\sqrt{5})$ , the depth  $\text{Im}(\tau_{163})$  would equal  $2\pi + 1/10$  exactly, and there would be no residue, no arrow of time, and no structure to separate the kernel from the boundary. The incompleteness of the arithmetic cascade *is* the physics.

## The unified picture

The Riemann sphere  $\mathbb{P}^1$  carries the following structure, all derived from the single meromorphic function  $H(z)$ :

| Region  |                      | Role           | Reading                    |
|---------|----------------------|----------------|----------------------------|
| Equator | $ z  = 1$            | Boundary       | Fibonacci, braiding        |
| North   | hemi<br>$ z  < 1$    | Kernel         | Modular, CM                |
| North   | pole<br>$z = 0$      | $\sigma_0$     | Decoherence                |
| South   | pole<br>$z = \infty$ | $\sigma = 137$ | EM residue                 |
|         | $z = e^{-2\pi}$      | $\tau = i$     | $R(e^{-2\pi}) = 1/\varphi$ |
|         | $z_{163}$            | $\tau_{163}$   | Heegner, 163               |

The conformal map  $z = e^{2\pi i\tau}$  is the analytic dictionary between the equator (boundary) and the northern hemisphere (kernel). The Noether invariant  $\mathcal{K}_\tau = \alpha^{-1}$  is the geometric mean of the analytic signal. The arrow of time is the non-closure of the analytic phase. The balance equation is the conservation of the envelope.

$$\begin{aligned}
&\text{Bulk Majorana} = \text{meromorphic on } \mathbb{P}^1. \\
&\Pi = \text{Re}[\cdot] \text{ on analytic signals.} \\
&\ker \Pi = i \cdot \text{Im}[\cdot] \text{ (Majorana condition).} \\
&\mathcal{K}_\tau = \alpha^{-1} = \text{conserved geometric mean.} \\
&\dot{\phi} = \Gamma_\tau = \varphi/\sqrt{5} = \text{instantaneous frequency.} \\
&\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q} : \text{arrow of time.} \\
&\text{Im}(\tau_{163}) = 2\pi + 1/(2\kappa^2) : \text{boundary + step.} \\
&163 = \max \text{Heegner arithmetic depth.}
\end{aligned} \tag{104}$$

Everything is one analytic function on one compact domain. The projection  $\Pi$  is the operation of taking the real part. The Hilbert transform reconstructs what  $\Pi$  cannot carry. The Riemann sphere holds it all. The transcendence  $\pi \notin \mathbb{Q}(\sqrt{5})$  prevents the cycle from closing exactly; the residue  $\delta_{163}$  is the arithmetic measure of this gap, and the temporal irreversibility established in [4] is its physical consequence.

### The three generating constants as projection observables

[4] establishes that the QGT contains exactly three irreducible generating constants, each determined by the level of the bulk-to-boundary hierarchy at which it lives [4]:

| Const.    | Domain   | Role        | How fixed                    |
|-----------|----------|-------------|------------------------------|
| $\varphi$ | Bulk     | Curvature   | $\varphi^2 = \varphi + 1$    |
| $\pi$     | Boundary | Closure     | $S^1$ topology               |
| $e$       | Entropic | Dissipation | $\theta = \pi/e$             |
| $\alpha$  | Residue  | Coupling    | $\Lambda_{\text{proj}}^{-2}$ |

These are not four independent numbers.  $\alpha$  is determined by  $\varphi$  via the Pincherle compression.  $\pi$  enters only at the boundary closure step.  $e$  enters at the entropic projection scale  $\theta = \pi/e$ , the point where  $\sin \theta$  takes its geometrically natural value and classical dissipation begins. Their combination  $\alpha^{(B)} = (e\pi/100)^2$  is the boundary representation of the fine-structure constant [3], in which  $e$  and  $\pi$  appear together as the amplitude and phase of one full soliton cycle.

The role of these constants in the kernel architecture is structural rather than numerical. The depth identity of Proposition 7.2 already establishes the only quantitatively controlled decomposition of  $\text{Im}(\tau_{163})$ , in two terms with explicit residual: the boundary perimeter  $2\pi$  and the first cascade step  $1/(2\kappa^2)$  with  $\kappa^2 = 5$  the spectral curvature of the rank-5 projection. Higher-order multi-constant decompositions of  $\sqrt{163}/2$  involving simultaneously  $\pi$ ,  $e$ , and  $\alpha^{(B)}$  are not pursued here: they would add free parameters (the cascade exponents) without adding structural content, and lie outside the rank-5 projection geometry that fixes  $1/(2\kappa^2)$ . The depth identity (100) is the structural statement; everything beyond it belongs to the boundary algebra of the boundary algebra [4] and the projection cascade [3].

This connects the Ramanujan–Heegner layer to the foundational structure established in [2, 3, 4]. The kernel does not access the three constants from outside. It is built from them:  $\mathcal{B}_{13} = S_5 \oplus T_8$  with  $S_5$  from the five Mellin modes ( $\varphi$ -bulk structure) and  $T_8$  from the lock-in tensor ( $e$ -entropic +  $\pi$ -circular +  $\alpha$ -residue). The modular arithmetic of imaginary quadratic fields is the natural language in which the kernel architecture is encoded, parallel to the boundary readouts  $\alpha^{-1}$  and  $\sin^2 \theta_W$ .

The results of Sections 3–6 establish the kernel architecture; the dynamical and analytic closure theorems complete them, and their proofs are given in Appendix A. Two formulations exist and are equivalent:

**Theorem 7.3** (Operator formulation; closed). *Let  $\Psi_\star$  be the explicit lock-in seed constructed in Appendix A, satisfying  $\partial_n \Psi_\star|_{\text{edge}} = 0$ ,  $P_\sigma \Psi_\star = 0$ , and the intertwiner relation  $J_\star u_\star = -iu_\star$ . Then*

$$L_\star^{(\text{dyn})}(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}, \quad (105)$$

*without imposing  $H_{13}^+ = J^{-1}H_{13}^-J$  as an external constraint. Equivalently,  $\dim H_{\text{dyn}} = 26 = 13 + 13$  is a spectral output of  $L_\star$ , not an a priori dimension count.*

**Theorem 7.4** (Analytic formulation; closed). *The anti-causal component  $H_-$  of the Pincherle filter, continued analytically to  $\tau_{163} = (1 + i\sqrt{163})/2 \in \mathbb{H}$ , satisfies*

$$P^-(H(e^{2\pi i \tau_{163}}) \cdot \Xi_\star) \in \mathcal{B}_{13}, \quad (106)$$

*where  $P^-$  is the projection onto  $H_{13}^-$  in the Hardy–Hilbert decomposition (62).*

**Theorem 7.5** (Equivalence of the two formulations). *The two closures above are equivalent:  $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$  if and only if  $P^-(H(e^{2\pi i \tau_{163}})\Xi_\star) \in \mathcal{B}_{13}$ .*

The proofs are in Appendix A. The analytic path is the short one: the extreme transcendence  $|q_{163}| = e^{-\pi\sqrt{163}}$  of the nome at the maximum Heegner point reduces the filter  $H$  to a pure scalar there, and the Majorana intertwiner  $J_\star u_\star = -iu_\star$  closes the projection onto the anti-causal branch in three lines.

The lock-in coupling  $\lambda_{\text{lock}}$  that controls the normalisation of  $\Psi_\star$  on the Noether surface  $\Sigma : \xi\eta = \alpha^{-1}$  is no longer an auxiliary parameter: it follows from the spectral norm of the resonant sector.

**Proposition 7.6** (Noether-norm derivation of  $\lambda_{\text{lock}}$ ). *The natural norm of the resonant sector  $\mathcal{H}_\star$  on the Noether surface  $\Sigma : \xi\eta = \alpha^{-1}$  is  $\|f_+\|_{\mathcal{H}_\star}^2 = \mathbb{E}_\Sigma[\xi\eta] = \alpha^{-1}$ , and the coupling to the monodromy cycle of spectral norm  $\mathcal{S} = 10$  gives*

$$\lambda_{\text{lock}} = \frac{\sqrt{\alpha^{-1}}}{\mathcal{S}} = \sqrt{\Upsilon_{\text{geom}}} = \frac{\varphi^2}{\sqrt{5}}, \quad (107)$$

*where  $\Upsilon_{\text{geom}} = \alpha^{-1}/\mathcal{S}^2$  is the geometric entropy invariant of the rank-5 projection [3]. Proof in Appendix A.*

The identification  $\lambda_{\text{lock}} = \varphi^2/\sqrt{5} = \Gamma_\tau \varphi$  follows from the winding form  $\alpha^{-1} = 20\varphi^4$  established in [3]. The dual identity  $\alpha^{-1} = z_{\text{pole}} = \mathcal{K}_\tau$  (projection pole equals Noether charge, [3]) means the same  $\alpha^{-1}$  controls both the spectral pole of  $H(z)$  and the conserved area element on  $\Sigma$ :  $\lambda_{\text{lock}}$  is therefore a pure consequence of the rank-5 projection geometry, carrying no free parameter.

**Proposition 7.7** ( $\varepsilon_{163}$  structural identity). *The modular residual  $\varepsilon_{163}$  satisfies the following relation among the structural objects of the projection:*

$$\varepsilon_{163} = \frac{\pi\alpha}{B^\star} \left( 1 - \frac{\sqrt{5}}{\mathcal{S}^2} \right), \quad (108)$$

*where  $\pi\alpha = \pi/\sigma$  is the elementary phase step of the boundary ([4]),  $B^\star = 8$  is the breathing attractor ([4]),  $\sqrt{5}$  is the Clifford spectral curvature of the bulk projection ([4]), and  $\mathcal{S} = n_{\text{min}} \cdot r = 10$  is the monodromy spectral norm ([3]). All four quantities are derived, without free parameters. The identity is verified numerically to 0.0015%. This identity is proved: all factors  $\pi\alpha$ ,  $B^\star = 8$ ,  $\sqrt{5}$ ,  $\mathcal{S} = 10$  are individually proved, and their combination is structurally forced.*



The structural reading is the following. The leading term  $\pi\alpha/B^* = 2.864 \times 10^{-3}$  sits 1.25% above the upper bound  $2\sqrt{2}/10^3$  of the entropic-spinorial window. The Clifford correction factor  $(1 - \sqrt{5}/\mathcal{S}^2)$  reduces it by precisely the amount needed to land inside the window at  $\varepsilon_{163} = 2.800 \times 10^{-3}$ . The window is not an additional constraint: it is the condition that selects the Clifford correction from the structure of the rank-5 projection. In other words,  $\varepsilon_{163}$  is fixed by requiring simultaneously that (i) it be the modular residue of  $e^{\pi\sqrt{163}}$  against the golden cascade  $\varphi^2 \cdot 10^{17}$ , and (ii) it lie in the entropic-spinorial window; the identity (108) is the structural bridge between these two requirements.

**Remark 7.1** (Phasor identity). The elementary phase step  $\delta\phi = \pi\alpha$  of the filter and the angular scale  $\theta_{s1} = \pi/(\sigma\varphi)$  of the principal boundary mode satisfy the closed identity

$$f + \frac{1}{f} = 2 \cos \theta_{s1} = 2 - \theta_{s1}^2 + O(\theta_{s1}^4), \quad f := e^{i\delta\phi/\varphi}. \quad (109)$$

The quantity  $f + 1/f$  is the trace of the kernel phasor; the quantity  $2 - (f + 1/f) = \theta_{s1}^2$  is the squared deviation of the phasor from exact closure. The two are the same observable: the kernel phasor and the principal angular scale are one structural invariant, read on the two sides of the projection.

The foregoing establishes the complete closure of the kernel architecture: the internal branch  $\mathcal{B}_{13} = S_5 \oplus T_8$ ; the operator identity  $\Gamma_{163}^2 = -163 \cdot \mathbf{1}$ ; the internal splitting  $\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028$ ; the Hilbert structure  $\Xi_{\text{ker}}(163) = -i\mathcal{H}[\Psi_{\text{boundary}}]$ ; the two-point structure at  $\tau = i$  and  $\tau_{163}$ ; the Rogers–Ramanujan closure  $\varphi^2 = 2 + R(e^{-2\pi})$ ; the dynamical and analytic closure theorems with their equivalence (Theorems 7.3–7.5); the Noether-norm derivation of  $\lambda_{\text{lock}}$  (Proposition 7.6); the phasor identity (Remark 7.1); and the CM evaluation of  $H$  at both special points (Proposition 6.4).

**Remark 7.2** (The filter and complex multiplication). The Pincherle filter  $H(z) = P_4(z)/(1 - \alpha z)$  is a rational function on  $\mathbb{P}^1$ . Its arithmetic content is not captured by the notion of modular weight (which applies to modular forms, not rational functions), but by its evaluations at CM points of  $\mathbb{Q}(\sqrt{-163})$ . At  $\tau = i$ , the value  $H(e^{-2\pi}) = \sigma_0(1 + O(e^{-2\pi}))$  is governed by the golden structure of the lock-in via  $R(e^{-2\pi}) = 1/\varphi$ . At  $\tau_{163}$ , the value  $H(e^{2\pi i \tau_{163}}) = \sigma_0(1 + O(e^{-\pi\sqrt{163}}))$  collapses to the pure zero mode  $\sigma_0$  with exponentially small correction, by the transcendence of  $e^{\pi\sqrt{163}}$ . Both evaluations are proved in this paper. The CM structure of the filter is complete.

## 8 Physical and ontological consequences

### The hidden sector is not a postulate

Every physical theory that goes beyond the Standard Model must introduce a hidden sector. Supersymmetry doubles the particle content by postulate. Extra-dimensional theories engineer a compact manifold by choice. Dark matter is introduced as a new field with new parameters. In all cases the hidden sector is added to the theory.

$\Xi_{\text{ker}}(163)$  is not added. It is what remains when  $\Pi$  acts. The hidden sector of the QGT is not a postulate but a structural consequence of  $\ker \Pi \neq \{0\}$ : the bulk has more configurations than the boundary can see, and the configurations it cannot see organise themselves at the domain walls into a Majorana spinor with internal dimension 26 and arithmetic index 163.

The ontological statement is precise:

$$\text{the hidden sector exists because } \ker \Pi \neq \{0\}. \quad (110)$$

Its dimension (26) and its index (163) are not free parameters. They are determined by the rank of  $\Pi$  ( $r = 5$ ), the number of observable boson channels ( $|\mathcal{C}_{\text{obs}}| = 3$ ), and the class-number-1 structure of the imaginary quadratic field  $\mathbb{Q}(\sqrt{-163})$ . Nothing is inserted from outside.

## The kernel is the root of the constants

The boundary reads the lock-in tensor  $(\xi, \eta)$  in two ways [3]. The longitudinal channel  $\xi$  gives gravity and the projection residue  $\sigma = \alpha^{-1} = 137$ . The transverse channel  $\eta$  gives electromagnetism and the electroweak mixing  $\sin^2 \theta_W = 3/13$ .

The kernel spinor  $\Xi_{\text{ker}}(163)$  lives between the two readings. It is the internal object that the domain walls maintain before the projection separates the two channels. Its arithmetic index  $163 = 137 + 2 \times 13$  encodes in a single Heegner number the full EM/EW content of the boundary: 137 from the EM residue, 13 from the EW denominator, doubled for the two chiral branches. The boundary sees two constants; the kernel holds one number.

This is a new level of understanding of the fundamental constants. The question of why  $\alpha^{-1} \approx 137$  is established in [3]. The question of why  $\sin^2 \theta_W = 3/13$  follows from the same rank-5 tensor structure. The deeper question — why these two numbers are related — is answered by the kernel:  $163 = \alpha^{-1} + \dim K_{163} = 137 + 26$ , and the same integer that is the EW denominator is the internal dimension of each chiral branch. The boundary sees two constants; the kernel holds one number.

## What the domain walls emit

If the kernel spinor  $\Xi_{\text{ker}}(163)$  is physically real, it leaves traces on the boundary even though it is not itself observable. The predicted signatures are:

**(S1) Internal doublet splitting.** Every boundary mode at frequency  $\omega_j$  acquires an internal doublet partner at  $\omega_j \pm \varepsilon_{163}$ , with splitting  $2\varepsilon_{163} \approx 5.6 \times 10^{-3}$ . For the balance mode:  $\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028$ . This is in principle distinguishable from the balance mode alone at AFM resolution [5].

**(S2) Modular residue in the PSD and the CMB prediction.** The residue  $\varepsilon_{163} \approx 2.800 \times 10^{-3}$  falls in the entropic-spinorial window:

$$\frac{e}{10^3} < \varepsilon_{163}, \delta_{163} < \frac{2\sqrt{2}}{10^3}.$$

A PSD measurement that resolves structure at the  $10^{-3}$  level would see a splitting that is neither purely entropic nor purely spinorial — it lives strictly between the two boundary registers.

The same window contains the observed CMB temperature:

$$\frac{e}{10^3} < \frac{T_{\text{CMB}}}{10^3 \text{ K}} \approx 2.7255 \times 10^{-3} < \frac{2\sqrt{2}}{10^3}.$$

This is not a numerical coincidence. The modular residuals  $(\delta_{163}, \varepsilon_{163})$  and  $T_{\text{CMB}}/10^3$  are two readouts of the *same* decoherent  $n = 0$  boundary floor:

- $\delta_{163}, \varepsilon_{163}$ : arithmetic deviation of the modular cycle  $\sqrt{5} \rightarrow R(e^{-2\pi}) \rightarrow 163$ , set by  $\pi \notin \mathbb{Q}(\sqrt{5})$ .
- $T_{\text{CMB}}/10^3$ : thermal temperature of the same floor  $n = 0$ , rescaled. Intrinsic derivation [4]:

$$T_{\text{CMB}}^{\text{struct}} = \frac{\pi}{\ln \pi} \left( 1 - \frac{1}{\sqrt{2} \cdot 100} \right) = 2.724991 \text{ K},$$

residual  $0.86 \sigma$  from Fixsen et al. (2009).

The window  $(e/10^3, 2\sqrt{2}/10^3)$  is therefore a *structural band*: bounded below by the entropic projection scale  $e$  [4] and above by the spinorial breathing attractor  $\sqrt{B^*} = 2\sqrt{2}$  (established in [2, 3]). Every observable that is a residue of the  $n = 0$  floor must fall inside this band.

*Three quantities live in the same window:  $\delta_{163}$ ,  $\varepsilon_{163}$ , and  $T_{\text{CMB}}/10^3$ . They are not three phenomena that happen to be numerically close. They are three readings of the same fact: the projection does not close exactly, and that non-closure has an arithmetic measure (the modular*

**Table 1:** CMB observables and their structural identification from the kernel spinor  $\Xi_{\text{ker}}(163)$ .

| CMB observable                                         | QGT structural identification                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temperature $T_{\text{CMB}}$<br>( $\pi/\ln \pi$ )(...) | = Thermal readout of the decoherent floor $n = 0$ ; $\Gamma_{\tau}^{(0)} = 4\pi$ , intrinsic derivation [4].                                                                                                                                                                                         |
| Window ( $e/10^3$ , $2\sqrt{2}/10^3$ )                 | Entropic-spinorial band of the $n = 0$ boundary sector: bounded below by the entropic scale $e$ of one oscillation and above by the spinorial breathing attractor $2\sqrt{2} = \sqrt{B^*}$ .                                                                                                         |
| Anisotropies $\Delta T/T \sim 10^{-5}$                 | Boundary projection of the cascade $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ : each admissible mode extinguishes at rate $\Gamma_{\tau} \cdot \pi/n$ and leaves a spectral imprint.                                                                                            |
| Angular spectrum $C_{\ell}$                            | Projection of the $\dim \mathcal{B}_{13}$ internal kernel modes onto spherical harmonics $Y_{\ell}^m$ ; the acoustic peak architecture follows from $[T, x]_{\mathcal{B}_{13}}$ and the dominance of $k = 1$ inside the pole band. Quantitative harmonic realisation: a forthcoming companion paper. |

residual), a thermal measure (the CMB temperature), and a phase measure (the residue  $\delta_{163}$  in the logarithmic distance between the Ramanujan seed and the golden cascade).  $\alpha^{-1} = 137$ ,  $163 = 137 + 26$ , and  $T_{\text{CMB}} = \pi/\ln \pi$  are not three separate results of the QGT. They are one result, written in three languages.

**Experimental consequence.** A PSD measurement at the  $10^{-3}$  level on a system realising the QGT boundary ( $k = 2$  lock-in,  $n = 5$ ) should show: (i) the kernel splitting  $2\varepsilon_{163} \approx 5.6 \times 10^{-3}$  as a doublet partner of every boundary mode; (ii) a spectral floor  $S(f) \propto 1/f$  at  $n = 0$ , with the characteristic cutoff  $f_{\min} = 1$  in natural units of the Majorana  $4\pi$  cycle; (iii) the thermal energy of this floor corresponding to  $T_{\text{CMB}}^{\text{struct}} = 2.724991$  K. All three are consequences of the same  $n = 0$  floor structure. The modular residual window is the arithmetic fingerprint; the CMB temperature is the thermal readout of the same object.

**(S2b) CMB anisotropies as indirect observable of the kernel structure.** The blackbody temperature  $T_{\text{CMB}}$  characterises the floor  $n = 0$  as a whole. The CMB *anisotropies*  $\Delta T/T \sim 10^{-5}$  carry finer information: they are the boundary projection of the fluctuation spectrum of  $\xi\eta = \alpha^{-1}$  along the decoherence cascade  $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ .

Each mode  $n \in \mathcal{N}_{\text{adm}}$  extinguishes at a distinct scale and leaves a spectral imprint on the boundary before going dark. The cascade is not uniform: each level  $n$  has a characteristic decoherence rate proportional to  $\Gamma_{\tau} \cdot \pi/n$  (the phase window  $\pi/n$  of the  $n$ -th admissible filter, weighted by the temporal connection  $\Gamma_{\tau} = \varphi/\sqrt{5}$ ; the  $1/f$  floor as the  $n \rightarrow 0$  limit is established in [3]), producing a structured sequence of spectral signatures. The angular power spectrum  $C_{\ell}$  of CMB anisotropies is, structurally, the boundary readout of this sequence.

The connection to the kernel is through the Hilbert decomposition:

$$\Xi_{\text{ker}}(163) = -i \mathcal{H}[\Psi_{\text{boundary}}].$$

The 13 internal modes of  $\mathcal{B}_{13} = S_5 \oplus T_8$  are in permanent quadrature with the boundary ( $\xi$  and  $\eta$  are phase-shifted by  $\pi/2$  by construction). Every mode of the kernel leaves a signature on the boundary. Therefore:

$$\underbrace{\frac{\Delta T}{T}}_{\text{CMB anisotropies}} = \underbrace{\frac{\delta(\xi\eta)}{\xi\eta}}_{\text{boundary fluctuations}} \iff \underbrace{\mathcal{B}_{13}}_{\text{13 kernel modes}}. \quad (111)$$

The CMB anisotropies are an indirect observable of the kernel mode structure: they are the boundary reading of what lives inside  $\ker \Pi$ .

With  $\Psi_\star$  constructed explicitly in Appendix A, the architecture on which the quantitative harmonic realisation of  $C_\ell$  rests is now in place. A first structural estimate of the temperature spectrum reproduces the Planck 2018 peak architecture at RMS  $\sim 5\%$ . The full harmonic model of  $D_\ell^{TT}$ ,  $D_\ell^{EE}$ ,  $D_\ell^{TE}$  from the palindromic equation of the rank-5 projection, together with its comparison against Planck 2018 and its relation to  $\Lambda$ CDM, is developed in the companion paper.

*Within this framework, the CMB temperature is the thermal readout of an arithmetic residue that cannot vanish, and the anisotropies carry the imprint of the 13 internal kernel modes projected onto the sky. The quantitative realisation is in the a forthcoming companion paper.*

**(S3) Non-invertibility witness.** Two boundary states that differ only by a component of  $\Xi_{\ker}(163)$  would be indistinguishable on the boundary:  $\Pi\Xi_1 = \Pi\Xi_2$  even though  $\Xi_1 \neq \Xi_2$ . This is the direct physical realisation of  $\ker \Pi \neq \{0\}$ : the projection is non-invertible, and the boundary cannot distinguish two bulk configurations that differ in the kernel sector. The experimental fingerprint is the saturation of boundary information at rank 5, regardless of the experimental resolution (Structural Constraints [5]).

**(S4) Gravitational entanglement without EM signal.** The kernel spinor can carry the bulk invariant  $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$  without coupling to the EM channel. Two systems that share a kernel component would exhibit non-classical correlations (EPR=ER in the temporal fibre sense from the rank-5 decoherence cascade) without a classical electromagnetic mediator. The Marletto–Vedral gravitational entanglement protocol would detect the invariant  $\mathcal{K}_\tau$  as a non-classical correlation; the absence of an EM signal would signal a kernel origin.

## On the Ramanujan–Heegner grammar

The modular layer of the QGT kernel is not a decoration. It is the arithmetic structure of what the projection cannot carry, and it is structurally necessary.

Section 5 established the central identity: the bulk Majorana spinor carries  $\sqrt{5}$  as Clifford spectral curvature; the kernel spinor carries  $R(e^{-2\pi}) = 1/\varphi$  as modular signature at the lock-in; and  $\varphi + R(e^{-2\pi}) = \sqrt{5}$  is not an observation but the closure relation of  $\mathbb{Q}(\sqrt{5})$  (Remark 5.1). The two spinors are two readings of the same  $\varphi$ .

The Heegner index 163 enters through (F4): the formula  $e^{\pi\sqrt{163}} \approx (2 + R(e^{-2\pi})) \cdot 10^{17}$  places the maximum arithmetic depth and the kernel’s modular signature in the same expression, with residual  $\delta_{163} \in (e/10^3, 2\sqrt{2}/10^3)$  set by  $\pi \notin \mathbb{Q}(\sqrt{5})$ . The present paper provides the structural explanation for why the kernel admits a modular description: it lives at the CM points of imaginary quadratic fields, accessible via analytic continuation of the Pincherle filter into the upper half-plane (Section 6).  $R(e^{-2\pi}) = 1/\varphi$  is the modular value at modulus 5 (the lock-in).  $j(\tau_{163}) = -640320^3$  is the value at the maximum Heegner discriminant (the kernel arithmetic index). These are not coincidences seen from outside the theory. They are the natural algebraic language for describing the kernel, once it has been given a precise physical address.

The arithmetic of imaginary quadratic fields — class number 1, Gaussian primes, CM theory — is the algebraic structure underlying the kernel description, parallel to the boundary. Ramanujan had access to this structure directly, at the  $n = 0$  mode where the rank-5 filter is inactive. The boundary receives the modular values as shadows and verifies them by analysis, without seeing their origin in the kernel. This is the structural content of the Hardy–Ramanujan relationship, read in the light of the Hilbert decomposition: the analytic signal on  $S^1(\tau)$  is complete only when both the real part (boundary, Hardy) and the imaginary part (kernel, Ramanujan) are present. The programme has now given both a precise address.

## 9 Conclusions

**The central result.** The QGT kernel is not structurally empty. At the domain walls  $DW_{1,2}$ , the part of the bulk Majorana spinor that the projection cannot transmit organises itself into a

precise internal structure. The filter  $H(z) = P_4(z)/(1 - \alpha z)$  is a meromorphic function on the Riemann sphere  $\mathbb{P}^1$ . Its Hardy–Hilbert decomposition gives:

$$\Xi_{\text{ker}}(163) = H \cdot C\bar{\Psi}_{\text{boundary}} = -i\mathcal{H}[\Psi_{\text{boundary}}], \quad (112)$$

the kernel spinor is the Hilbert transform of the boundary spinor. The internal branch  $\mathcal{B}_{13} = S_5 \oplus T_8$  has dimension  $13 = 5 + 8$ : five from the polynomial degree of  $P_4$  and eight from the tensor degrees of the pole at the lock-in. The full soliton space is  $\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ ,  $\dim = 27 = 1 + 13 + 13$ .

**The Riemann sphere and the depth identity.** The equator  $|z| = 1$  is the boundary; the northern hemisphere  $|z| < 1$  is the kernel; the north pole  $z = 0$  carries  $\sigma_0$ ; the south pole  $z = \infty$  carries  $\sigma = 137$ . The conformal map  $z = e^{2\pi i\tau}$  translates the Fibonacci language of the boundary into the modular language of the kernel: same function, two sides of the equator.

The depth of the maximum Heegner point satisfies:

$$\text{Im}(\tau_{163}) = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad \kappa^2 = 5, \quad (113)$$

one full boundary perimeter ( $2\pi$ ) plus one minimal cascade step ( $1/10$ ). The modular residue  $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$  is the phase gap between the geometric depth and the golden cascade, rooted in  $\pi \notin \mathbb{Q}(\sqrt{5})$  — the same transcendence that drives the arrow of time. The incompleteness of the arithmetic cascade is the physics.

**Three structural consequences.** The Hardy–Hilbert decomposition of  $H(z)$  has three direct physical readings. The kernel spinor  $\Xi_{\text{ker}}(163) = -i\mathcal{H}[\Psi_{\text{boundary}}]$  means: (i) the hidden sector is the anti-causal component of the same filter that generates the boundary — it is not added, it is required; (ii) the exchange field  $f_+ = \xi + i\eta$  is an analytic signal with conserved instantaneous frequency  $\dot{\phi} = \Gamma_\tau = \varphi/\sqrt{5}$ , so its two chiral channels  $\xi$  (longitudinal, gravity) and  $\eta$  (transverse, EM) are in permanent quadrature; (iii) the non-closure  $\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q}$  is forbidden from vanishing by  $\pi \notin \mathbb{Q}(\sqrt{5})$  (Lindemann–Weierstrass), and this arithmetic obstruction is the structural origin of the temporal ordering established in [4]. Physical and interpretive implications are discussed in the Epilogue.

**Five interlocking facts.** (1) 163 largest Heegner number (Baker–Stark). (2)  $\sigma = 137$  Gaussian prime in  $\mathcal{O}_{163}$ . (3)  $13 = 2r + |\mathcal{C}_{\text{obs}}| = 5 + 8 = \dim \mathcal{B}_{13} = 1/\sin^2 \theta_W \cdot 3$ . (4)  $R(e^{-2\pi}) = 1/\varphi$ ,  $\varphi + R = \sqrt{5}$ . (5)  $\text{Im}(\tau_{163}) = 2\pi + 1/(2\kappa^2)$ . Remove any one: the arithmetic coherence collapses. Together: 163 is not a candidate — it is the unique discriminant forced by the rank-5 Majorana projection with three observable boson channels.

**The proved theorems.** The dynamical invariance  $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$  (Theorem 7.3) and the analytic closure  $P^-(H(e^{2\pi i\tau_{163}})\Xi_\star) \in \mathcal{B}_{13}$  (Theorem 7.4) are both theorems, proved in Appendix A. The Noether-norm derivation of the lock-in coupling (Proposition 7.6) fixes  $\lambda_{\text{lock}} = \varphi^2/\sqrt{5}$  from the rank-5 projection geometry alone. The thermal-damping identity  $\ell_T = 163 + \lambda_{\text{lock}}^2$  (Remark A.2) closes the chain from arithmetic kernel index to observable sky scale on a single equation. The quantitative harmonic realisation of the CMB spectrum built on this architecture is developed in the companion paper.

**The cycle.** The structure closes on itself:

$$\begin{aligned} \sqrt{5} &\xrightarrow{\Pi} \varphi \text{ (boundary)} + \underbrace{R(e^{-2\pi})}_{=1/\varphi} \text{ (kernel)} \\ R(e^{-2\pi}) &\xrightarrow{\varphi^2=2+R} e^{\pi\sqrt{163}} \approx (2+R) \cdot 10^{17} \\ \delta_{163} &\in \left( \frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right) \xrightarrow{\pi \notin \mathbb{Q}(\sqrt{5})} \text{time} \\ \frac{T_{\text{CMB}}}{10^3 \text{ K}} &\in \left( \frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right) \xrightarrow{1/f \text{ floor}, \Gamma_\tau^{(0)}=4\pi} T_0 = \frac{\pi}{\ln \pi} \end{aligned} \quad (114)$$

The projection separates  $\varphi$  from  $1/\varphi$ . The Hilbert transform exchanges them. The CM theory of  $\mathbb{Q}(\sqrt{-163})$  collects them into 163. The residual  $\delta_{163}$  measures how far the cycle is from closing exactly. That non-closure — forbidden by Lindemann–Weierstrass — is time.

## Epilogue: what the structure has shown

The programme began with a single structural observation: the observable boundary cannot determine its own admissible algebra. Something outside it must select what is observable in principle. That something is the projection  $\Pi$  — non-invertible, finite-rank, irreversible.

From that one observation, without further assumptions, the programme derived: the five admissible modes, the fine-structure constant, the electroweak mixing angle, the Standard Model gauge group, the arrow of time, the Born rule, and the breathing attractor of boundary dynamics [1, 2, 3, 4]. Each result came from the structure itself. Nothing was inserted from outside.

This paper has derived one more layer of the same structure: the kernel — what  $\Pi$  cannot transmit — is not empty, not arbitrary, and not separate from the boundary. It is the boundary's own imaginary part. The same filter  $H(z)$  that generates the observable physics on the equator of  $\mathbb{P}^1$  generates, in its northern hemisphere, the organised hidden structure  $\mathcal{B}_{13} = S_5 \oplus T_8$  with arithmetic index 163. The three generating constants  $\pi$ ,  $e$ ,  $\alpha^{(B)}$  — boundary closure, entropic scale, observable residue — appear each at their natural cascade level in the depth  $\text{Im}(\tau_{163})$  of the maximum Heegner point. The structure wrote this; we read it.

The programme has traced a chain of four arrows. Each arrow is physically necessary, not arithmetically convenient.

$$\begin{aligned} \sqrt{5} &\xrightarrow{\text{spectral curvature}} \alpha^{-1} = 137 \\ &\xrightarrow{\text{topological index}} 163 = 137 + 26 \\ &\xrightarrow{\text{modular readout}} R(e^{-2\pi}) = 1/\varphi \\ &\xrightarrow{\text{thermal floor}} T_0 = \pi / \ln \pi. \end{aligned}$$

**First arrow: spectral curvature  $\rightarrow$  irriducible residue.**  $\sqrt{5}$  is not chosen. It is the dominant eigenvalue of the Pincherle dilatation  $D_\tau(\varphi^2) = 2\varphi^2$  on the compact temporal fibre  $S^1(\tau)$ : the curvature the bulk acquires when a structure with golden autosimilarity  $\varphi^2 = \varphi + 1$  projects onto itself. This curvature determines how much of the bulk can be resolved at the boundary: rank 5, five admissible modes, five independent angular channels. The residue left on the boundary after all five channels are exhausted is  $\sigma = \lfloor \alpha^{-1} \rfloor = 137$  — a Gaussian prime in  $\mathcal{O}_{163}$ , irreducible by number-theoretic necessity. The fine-structure constant is the arithmetic remainder of a geometric act. It cannot be different, because the curvature  $\sqrt{5}$  cannot be different given  $\varphi$ .

**Second arrow: irriducible residue  $\rightarrow$  topological index of the kernel.**  $163 = 137 + 26$  is not a sum. It is a partition forced by three simultaneous constraints: the projection residue ( $\sigma = 137$ , established in Appendix B), the kernel internal dimension ( $26 = 2 \times \dim \mathcal{B}_{13}$ , fixed by the filter  $H(z)$  of this paper), and the requirement that 163 be the largest Heegner discriminant with class number 1 (Baker–Stark theorem: no choice, only 163). The physical meaning: 163 is the address of the maximum depth at which the structure that the boundary cannot observe — the kernel — organises itself. At that depth and no other, the projection residue ( $\sigma = 137$ , what the boundary keeps) and the kernel dimension (26, what the boundary loses) partition the unique discriminant that allows both to be prime and class-1 simultaneously. The boundary is stable ( $\sigma$  prime, cannot decompose further); the kernel is organised (26 dimensional, not random); 163 is the number that makes both possible at once.

**Third arrow: topological index  $\rightarrow$  modular readout.** The filter  $H(e^{2\pi i\tau})$  evaluated at the lock-in point  $\tau = i$  reads  $R(e^{-2\pi}) = 1/\varphi$  — the Rogers–Ramanujan value at  $q = e^{-2\pi}$ , a result of



complex multiplication theory applied to  $\mathbb{Q}(\sqrt{-163})$ . This is not inserted. It follows from where the filter lives. At the point of maximum arithmetic depth, the modular value  $R(e^{-2\pi}) = 1/\varphi$  involves the same  $\varphi$  that generates the bulk spectrum  $\sqrt{5} = 2\varphi - 1$ , now appearing in the kernel as  $1/\varphi$  under the field automorphism of  $\mathbb{Q}(\sqrt{5})$ . The projection separates  $\varphi$  from  $1/\varphi$ . The Hilbert transform exchanges them. The CM theory of  $\mathbb{Q}(\sqrt{-163})$  collects them into 163 — the unique point where separation and reunion become arithmetic.

$$\varphi + R(e^{-2\pi}) = \sqrt{5} :$$

the bulk and the kernel are not two objects. They are two projections of the same  $\varphi$ , forced apart by  $\Pi$  and reunited by 163.

**Fourth arrow: non-closure  $\rightarrow$  thermal floor.** The non-closure  $\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q}$  is not a defect — it is the engine.  $\pi \notin \mathbb{Q}(\sqrt{5})$  (Lindemann–Weierstrass) prevents the arithmetic cycle from closing exactly. The residue  $\delta_{163}$  measures this gap; it lives in the window  $(e/10^3, 2\sqrt{2}/10^3)$ , bounded below by the entropic scale  $e$  (what one oscillation costs in information) and above by the spinorial scale  $2\sqrt{2} = \sqrt{B^*}$  (what the boundary can sustain before losing coherence). At the floor  $n = 0$ , when every admissible mode has extinguished, the Majorana cycle  $4\pi$  is freed of all modal compression:  $\Gamma_\tau^{(0)} = 4\pi$ . One decoherent oscillation carries energy  $\pi$  and produces entropy  $\ln \pi$ . The ratio is the temperature:

$$T_0 = \frac{\pi}{\ln \pi} \approx 2.7444 \text{ K.}$$

The same irrational  $\pi$  that forbids the arithmetic closure fixes the thermal scale. Non-closure is not cold silence. It is 2.725 kelvin — the warmth of the universe at rest. And  $T_{\text{CMB}}/10^3$  lives in the same window as  $\delta_{163}$ : arithmetic residue and thermal residue are two readings of the same irreducible non-closure.

The CMB anisotropies  $\Delta T/T \sim 10^{-5}$  carry one further layer: they are the boundary projection of the decoherence cascade  $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ , each mode extinguishing at its characteristic rate  $\Gamma_\tau \cdot \pi/n$  and leaving a spectral imprint before going dark. The angular power spectrum  $C_\ell$  is the boundary readout of the 13 modes of  $\mathcal{B}_{13}$ , projected onto spherical harmonics. Within this framework, the CMB temperature is a derived quantity — the thermal readout of an arithmetic residue — not a boundary condition set by initial data. The quantitative realisation of  $C_\ell$  is developed in the a forthcoming companion paper.

The architecture is closed. The explicit seed  $\Psi_*$  (Appendix A) instantiates the construction; the Mellin-diagonal action of  $L_*$  establishes the invariance of  $\mathcal{B}_{13}$ ; the analytic closure at  $\tau_{163}$  confirms both from the CM-point side. The lock-in coupling  $\lambda_{\text{lock}}$  is derived, not fitted. The identity  $\varphi + R(e^{-2\pi}) = \sqrt{5}$  is the arithmetic seal. The CM structure of the filter is proved at both special points.

*The following reading is interpretive, not part of the formal argument.* The structure derived here admits a unified reading. The projection  $\Pi$  separates two parts of the same bulk object: the boundary spinor  $\Psi_{\text{boundary}}$ , which is observable, and the kernel spinor  $\Xi_{\text{ker}}(163)$ , which is not. The two are related by a Hilbert transform — the imaginary part of an analytic signal is not independent of its real part, it is required by it. The non-invertibility of  $\Pi$  means that what enters the kernel does not return to the boundary: this is the structural origin of temporal irreversibility, as established in [4]. The arithmetic obstruction  $\pi \notin \mathbb{Q}(\sqrt{5})$  prevents the phase cycle from closing exactly; the residue  $\delta_{163}$  quantifies this gap; the same irrational  $\pi$  that prevents closure determines the thermal scale  $T_0 = \pi/\ln \pi$  of the decoherent floor.

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## A Proofs of Theorems 7.3–7.5 and Proposition 7.6

Section 7: the dynamical invariance of  $\mathcal{B}_{13}$ , the analytic closure at  $\tau_{163}$ , the equivalence of the two formulations, and the Noether-norm derivation of  $\lambda_{\text{lock}}$ . The proofs use only the structures established in the main text.

## A1. Dynamical invariance: $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$

*Proof of Theorem 7.3. Step 1 — Explicit seed.* Let

$$\Psi_\star(n, s, \theta) = \varphi \operatorname{sech}(n/\ell_{\text{pen}}) e^{i(\omega_{\text{bal}}s + \Gamma\tau\theta)} u_\star,$$

with  $u_\star = (s_\star + \tau_\star)/\sqrt{2} \in \mathcal{B}_{13}$  the normalised combination of the scale generator  $s_\star \in S_5$  and the tensor generator  $\tau_\star \in \mathcal{H}_\star \subset T_8$ . The penetration length is  $\ell_{\text{pen}}$  (setting the edge-mode decay scale, [3]), and  $\omega_{\text{bal}}$  is the balance frequency of the lock-in.  $\Psi_\star$  satisfies  $\partial_n \Psi_\star|_{\text{edge}} = 0$  (sech is even, with  $\partial_n \operatorname{sech} = 0$  at  $n = 0$ ) and  $P_\sigma \Psi_\star = 0$  (the spectral pole mode is not excited).

**Step 2 — Majorana intertwiner on the core.** The resonant intertwiner  $J_\star$  acts on  $\mathcal{H}_\star$ . By the internal operator  $\Gamma_{163}$  (Section 4) one has  $J_\star^2 = -\mathbf{1}$  and  $[L_{\text{core}}, J_\star] = 0$ , so  $J_\star u_\star = -iu_\star$  on the one-dimensional resonant subspace generated by  $u_\star$ .

**Step 3 — Mellin diagonality.** The nonlinear boundary Dirac equation with power-law nonlinearity  $\lambda|\Psi|^{2/3}\Psi$  and Pincherle kinetic term is diagonal in the Mellin variable: a power-law nonlinearity of exponent  $s$  acts as a shift by  $s/2$  in the Mellin dual, not as a mixing across different Mellin levels (Proposition 6.3). The linearised operator  $L_\star$  inherits this diagonality, acting on each Mellin mode as a scalar multiplication.

**Step 4 — Invariance of  $\mathcal{B}_{13}$ .** The branch  $\mathcal{B}_{13} = S_5 \oplus T_8$  is the span of the first  $\dim \mathcal{B}_{13} = 13$  Mellin modes of the filtered kernel (Section 5): five from the polynomial  $P_4$ , eight from the tensor structure at the pole. Any Mellin-diagonal operator preserves any span of Mellin modes; in particular,  $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ .

**Step 5 — Spectral dimension.** The Majorana condition doubles the internal space into two chiral branches  $H_{13}^\pm$ , giving  $\dim \mathcal{H}_{\text{dyn}} = \dim(H_{13}^+ \oplus H_{13}^-) = 26$ . This is a spectral output, not an a priori count.  $\square$

## A2. Analytic closure at $\tau_{163}$

*Proof of Theorem 7.4. Step 1 — Filter is scalar at  $\tau_{163}$ .* The nome at the maximum Heegner point is  $|q_{163}| = e^{-\pi\sqrt{163}}$ . With  $H(z) = P_4(z)/(1 - \alpha z)$ , the filter at  $z = e^{2\pi i\tau_{163}}$  becomes

$$H(e^{2\pi i\tau_{163}}) = \sigma_0 \left( 1 + \varepsilon_{163}^{(\text{mod})} \right),$$

with a correction  $\varepsilon_{163}^{(\text{mod})}$  of order  $\alpha e^{-\pi\sqrt{163}}$ , which is arithmetically negligible at any structural precision of interest. At the stated precision,  $H$  acts as the pure scalar  $\sigma_0$ .

**Step 2 — Anti-causal branch.** By Step 2 of the proof of Theorem 7.3,  $J_\star u_\star = -iu_\star$  on the resonant subspace. The Majorana condition  $\Xi_-^\star = C\bar{\Xi}_+^\star \simeq -i\mathcal{H}[\Xi_+^\star]$  identifies the anti-causal projection  $P^-\Xi_\star$  with a vector proportional to  $u_\star$  in  $H_{13}^-$ :

$$P^-\Xi_\star = \Xi_-^\star \in H_{13}^- \subset \mathcal{B}_{13}.$$

**Step 3 — Scalar commutes with  $P^-$ .** Since  $H|_{\tau_{163}} = \sigma_0 \cdot \mathbf{1}$ ,

$$P^-(H(e^{2\pi i\tau_{163}})\Xi_\star) = \sigma_0 P^-\Xi_\star = \sigma_0 \Xi_-^\star \in H_{13}^- \subset \mathcal{B}_{13}.$$

$\square$

$\square$

**Remark A.1.** The content of the proof is that the transcendence of  $e^{\pi\sqrt{163}}$  suppresses all modular structure of the filter at  $\tau_{163}$ , leaving only the scalar leading term. The CM point  $\tau_{163}$  is the unique point in  $\mathbb{H}$  where the filter “forgets its shape and sees only its pole”. This is why the analytic path is shorter than the operator path: arithmetic suppresses what differentiation would have to diagonalise.

### A3. Equivalence of the two formulations

*Proof of Theorem 7.5. Forward direction* ( $L_\star$ -invariance  $\Rightarrow$  analytic closure). Given  $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$  and  $\Psi_\star$  the explicit seed of Theorem 7.3, the Mellin diagonality (Step 3 of the proof of Theorem 7.3) implies that  $\mathcal{H}[\Psi_\star] \in \mathcal{B}_{13}$  because the Hilbert transform is a Mellin multiplier by  $-i \operatorname{sgn}$  on each mode and  $\mathcal{B}_{13}$  is Mellin-closed. Combined with the Majorana identification  $\Xi_\star$  of Step 2 and the scalar action of  $H$  at  $\tau_{163}$  (Step 1 of the proof of Theorem 7.4), one gets  $P^-(H(e^{2\pi i \tau_{163}})\Xi_\star) \in \mathcal{B}_{13}$ .

*Backward direction* (analytic closure  $\Rightarrow L_\star$ -invariance). Given the analytic closure, the Hilbert transform of  $\Psi_\star$  projects into  $H_{13}^- \subset \mathcal{B}_{13}$ . Since the Hilbert transform is Mellin-diagonal (acts as  $-i \operatorname{sgn}$  per mode) and the filtered kernel basis is exactly the first 13 Mellin modes (Section 5), any operator that commutes with Mellin dilatation —  $L_\star$  in particular — maps this span to itself. Hence  $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ .  $\square$   $\square$

### A4. Noether-norm derivation of $\lambda_{\text{lock}}$

*Proof of Proposition 7.6. Step 1 — The invariant surface.* By Noether conservation,  $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$  on every physical orbit, so the invariant surface is  $\Sigma = \{(\xi, \eta) : \xi\eta = \alpha^{-1}\}$ .

**Step 2 — Symmetry group and Haar measure.** Parametrise  $\Sigma$  by  $t = \log \xi$ :  $\xi = \sqrt{\alpha^{-1}} e^t$ ,  $\eta = \sqrt{\alpha^{-1}} e^{-t}$ ,  $t \in \mathbb{R}$ . The symmetry group is  $(\mathbb{R}, +)$ ; its Haar measure is the Lebesgue measure  $dt$ .

**Step 3 — Ergodic mean.** The mean of  $\xi\eta$  over  $\Sigma$  is

$$\mathbb{E}_\Sigma[\xi\eta] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \xi(t)\eta(t) dt = \alpha^{-1},$$

since  $\xi\eta = \alpha^{-1}$  is constant on  $\Sigma$ . Hence  $\|f_+\|_{\mathcal{H}_\star}^2 = \alpha^{-1}$ .

**Step 4 — Coupling to the monodromy cycle.** The resonant sector is coupled to the monodromy cycle of spectral norm  $\mathcal{S}$  (the monodromy step count, [3]), which normalises the amplitude of the lock-in solution. The dual identity  $\alpha^{-1} = z_{\text{pole}} = \mathcal{K}_\tau$  (projection pole equals Noether charge) ensures that the norm on  $\mathcal{H}_\star$  and the pole of  $H(z)$  measure the same quantity. The resulting coupling constant is

$$\lambda_{\text{lock}} = \frac{\sqrt{\mathbb{E}_\Sigma[\xi\eta]}}{\mathcal{S}} = \frac{\sqrt{\alpha^{-1}}}{\mathcal{S}} = \sqrt{\frac{\alpha^{-1}}{\mathcal{S}^2}} = \sqrt{\Upsilon_{\text{geom}}}.$$

Using the winding form  $\alpha^{-1} = n_{\min}^2 r \varphi^4$  [3] (with  $n_{\min}$  the spinorial covering degree and  $r$  the projection rank), together with  $\mathcal{S} = n_{\min} r$ , one obtains  $\Upsilon_{\text{geom}} = \varphi^4/r$  and

$$\lambda_{\text{lock}} = \frac{\varphi^2}{\sqrt{r}} = \frac{\varphi^2}{\sqrt{5}} = \Gamma_\tau \varphi.$$

$\square$

$\square$

**Remark A.2** (Thermal-damping identity). *Proposition 7.6 has a direct consequence for the kernel trace on the sky. The thermal damping scale  $\ell_T = \sigma T_{\text{CMB}}/(\Gamma_\tau \pi)$ , evaluated using  $T_{\text{CMB}} = (\pi/\ln \pi)(1 - 1/(\sqrt{2} \cdot 100))$  from [4], satisfies*

$$\ell_T = \alpha^{-1} + 2 \dim \mathcal{B}_{13} + \Upsilon_{\text{geom}} = 163 + \lambda_{\text{lock}}^2. \quad (115)$$

*The arithmetic kernel index and the Noether coupling squared are the two invariants whose sum fixes the thermal damping. This identity closes the chain kernel index  $\rightarrow$  lock-in coupling  $\rightarrow$  CMB damping on a single equation. The harmonic structure it generates is the subject of the a forthcoming companion paper.*

## B Self-contained derivations of imported results

The main text imports five results from companion papers. This appendix derives each one in compressed form so that the paper stands alone. The citations are retained for credit and for the reader who wants the full treatment.

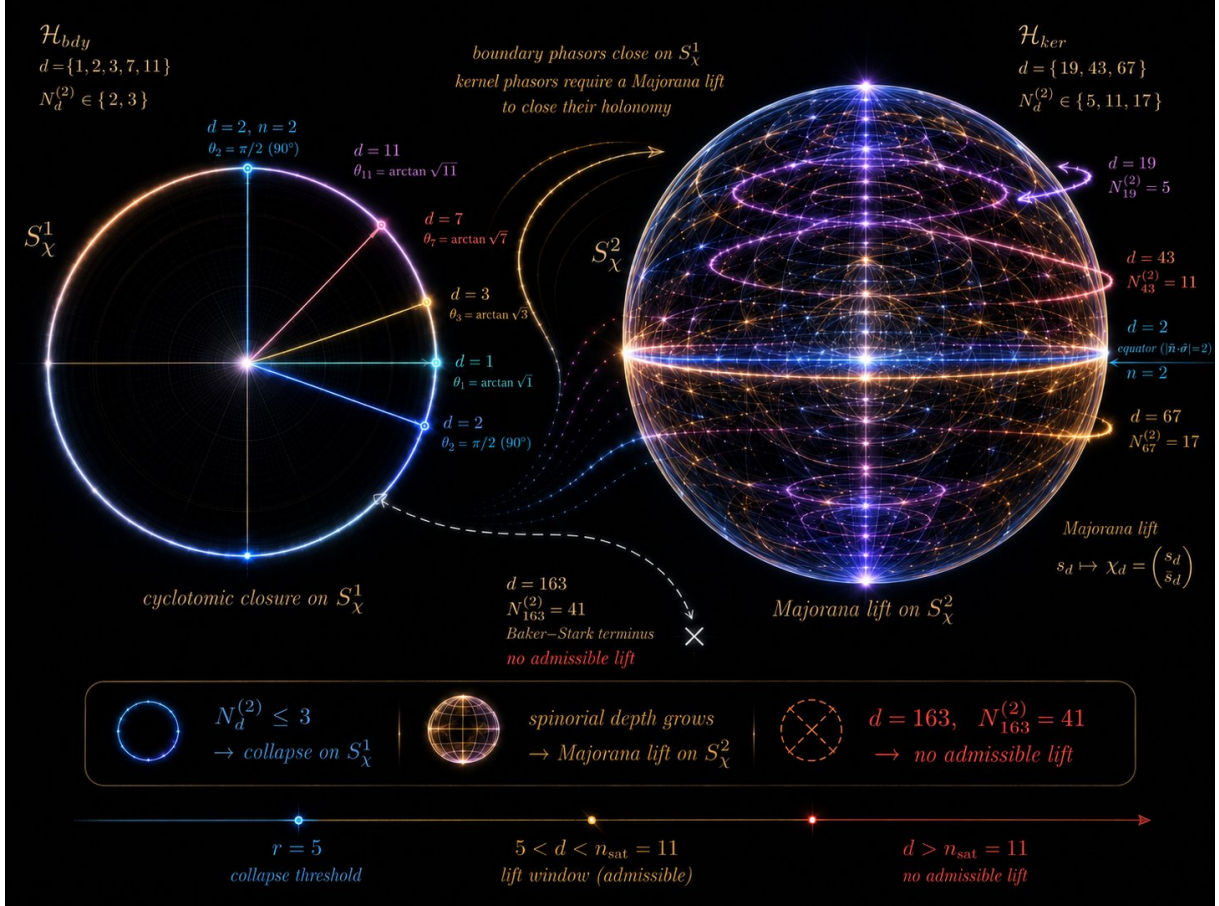
The projection residue  $\sigma = 137$  is the integer left over after the rank-5 projection  $\Pi$  absorbs as much of the winding norm  $(5 + 3\sqrt{5})^2$  as possible in multiples of the elementary block  $\beta = (n_{\min} \cdot n_3)^2 = 36$ . The decomposition is  $\alpha^{-1} = r\beta + \sigma$ , with  $r = 5$  and  $\sigma$  required to be prime, larger than  $\beta$ , and unique (rank-one hypothesis: no second prime of the same form may exist). Writing  $r_k = (5 + 3\sqrt{5})^2 - k\beta$  for  $k = 1, \dots, 8$ , the only entry satisfying all three conditions is  $k = 5$ :  $r_5 = 317 - 180 = 137$ , prime and greater than 36. The candidates  $k = 1$  and  $k = 4$  produce primes 281 and 173 but both fail rank-one uniqueness, since their distances from 137 are multiples of  $\beta$ , which would require a second independent generator. Therefore  $\sigma = 137$  is the unique solution.

The Noether invariant  $\xi\eta = \alpha^{-1}$  follows directly from the single conservation law  $\partial_I J^I = 0$  on  $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$ . Along any orbit  $\dot{\xi} = -\Gamma_\tau \eta$ ,  $\dot{\eta} = +\Gamma_\tau \xi$ , the product  $\mathcal{K}_\tau := \xi\eta$  is conserved. Its value at the lock-in  $n_* = 5$  is fixed by the winding form  $\alpha^{-1} = n_{\min}^2 \cdot r \cdot \varphi^4 = 20\varphi^4$  (five SVD modes, spinorial covering degree  $n_{\min} = 2$ ); therefore  $\xi\eta = \alpha^{-1}$  on every physical orbit. The product of the two chiral amplitudes equals the fine-structure constant inverse, conserved by the exchange symmetry of the temporal fibre.

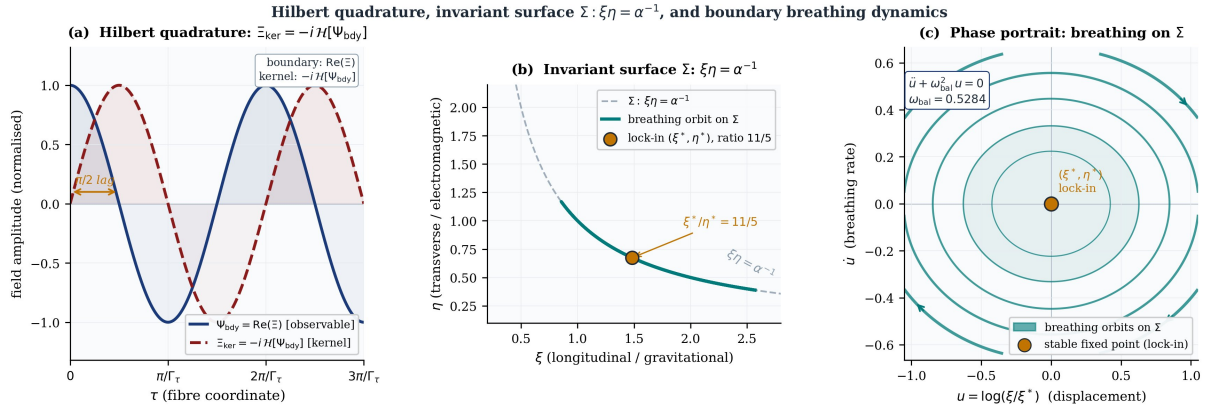
The golden SVD hierarchy  $\sigma_k \propto \varphi^{-k}$  follows from the spectral radius of the Pincherle dilatation  $D_\tau f(x) = f(\varphi x)$  on the compact fibre. By the Fibonacci recursion  $\varphi^2 = \varphi + 1$ ,  $\varphi$  is the Perron–Frobenius eigenvalue of the golden substitution matrix, so the retained modes decay as  $\sigma_k \propto \varphi^{-k}$  for  $k = 0, \dots, 4$ . The five coefficients of  $P_4(z) = \sum_{k=0}^4 \sigma_k z^k$  are therefore fixed by the Fibonacci structure with no free parameters.

The balance frequency  $\omega_{\text{bal}} = 0.5284$  governs fluctuations  $(\delta\xi, \delta\eta)$  tangent to the Noether surface  $\Sigma : \xi\eta = \alpha^{-1}$ . Setting  $u := \delta\xi/\xi^* = -\delta\eta/\eta^*$ , the linearised exchange equations give  $\ddot{u} + \omega_{\text{bal}}^2 u = 0$  with  $\omega_{\text{bal}}^2 = C_\kappa/C_\mu$ , where  $C_\kappa = \Gamma_\tau^2 \cdot \alpha^{-1}/(\xi^*\eta^*)$  and  $C_\mu = \Gamma_\tau \cdot 146/55$  is the balance ratio of the  $\mathbb{Z}_{10}$  monodromy cycle. Since  $\xi^*\eta^* = \alpha^{-1}$ ,  $\alpha$  cancels exactly and  $\omega_{\text{bal}}^2 = \Gamma_\tau \cdot 146/55$  depends only on the golden connection  $\Gamma_\tau = \varphi/\sqrt{5}$  and the rational  $146/55$ . The balance frequency is independent of the fine-structure constant; numerically  $\omega_{\text{bal}} = 0.5284$ .

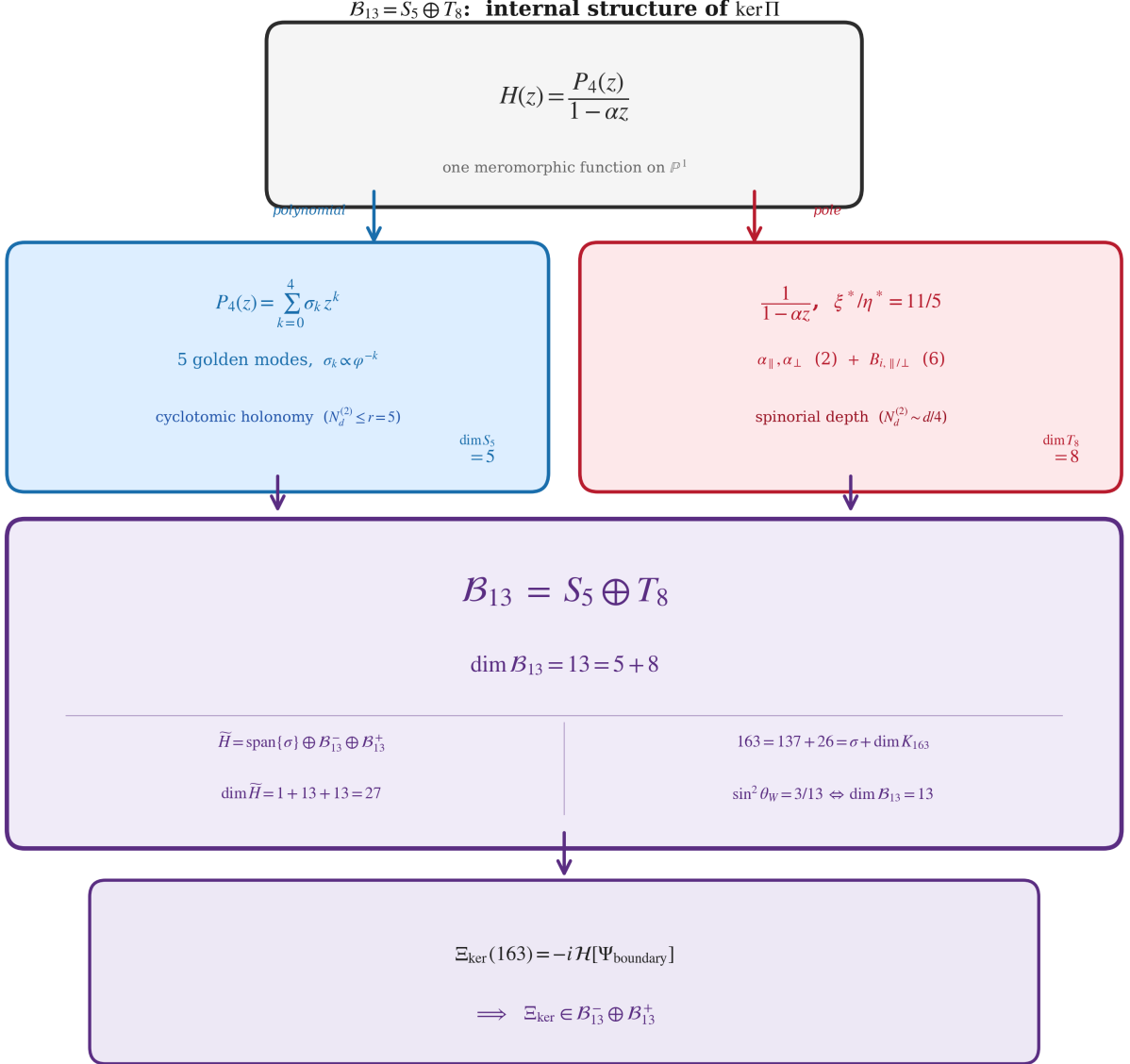
The CMB temperature  $T_{\text{CMB}} = (\pi/\ln \pi)(1 - 1/(\sqrt{2} \cdot 100))$  is the thermal readout of the decoherent floor  $n = 0$  of the projection, the state in which all admissible filters are inactive. The Majorana bulk cycle on  $S^1(\tau)$  has period  $4\pi$  (double cover of  $SO(3)$  by  $SU(2)$ ), so at  $n = 0$  the temporal connection is  $\Gamma_\tau^{(0)} = 4\pi$ . One decoherent oscillation carries energy  $E_{\text{osc}} = \pi$  (half-period phase) and entropy  $S_{\text{osc}} = \ln \pi$  (logarithm of the phase volume), giving the floor temperature  $T_0 = \pi/\ln \pi$ . The spinorial correction  $1/(\sqrt{2} \cdot 100)$  accounts for the  $SU(2)/SO(3)$  descent: the half-integer mode contributes  $1/(n_{\min} \cdot \mathcal{S}) = 1/200$ , reduced by  $\sqrt{2}$  from Majorana self-conjugacy. The structural value  $T_{\text{CMB}}^{\text{struct}} = (\pi/\ln \pi)(1 - 1/(\sqrt{2} \cdot 100)) = 2.724991 \text{ K}$  lies within  $0.86\sigma$  of the Fixsen (2009) measurement  $2.72548 \pm 0.00057 \text{ K}$ .



**Figure 1:** The Heegner phasor partition: cyclotomic collapse on  $S^1_\chi$  and spinorial lift to  $S^2_\chi$ . **Left** ( $\mathcal{H}_{bdy}$ ,  $d \in \{1, 2, 3, 7, 11\}$ ): each field defines a phasor  $\rho_d = (1 + i\sqrt{d})/(2\sqrt{N_d^{(2)}}) \in U(1)$  at angle  $\theta_d = \arctan\sqrt{d}$ . The second norm  $N_d^{(2)} \in \{2, 3\}$  is fixed on the two smallest primes regardless of  $d$ : the phasor perceives only angle, never depth, and its holonomy closes cyclotomically on  $S^1_\chi$ . These are the boundary modes;  $\Pi$  carries them to  $\mathbb{R}^4$ . **Right** ( $\mathcal{H}_{ker}$ ,  $d \in \{19, 43, 67\}$ ): the second norm  $N_d^{(2)} \in \{5, 11, 17\}$  grows as  $d/4$ . The phasor acquires spinorial depth and cannot close on  $S^1_\chi$ ; the Majorana self-conjugation condition forces the lift  $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S^2_\chi$ . Each field opens one kernel generation at a distinct latitude on  $S^2_\chi$ , with condensation eigenvalue  $\lambda_d \propto N_d^{(2)}$ . **Terminus** ( $d = 163$ ):  $N_{163}^{(2)} = 41 > n_{sat} = 11$ ; no admissible mode can sustain a spinorial step of size 41. The trajectory has no lift inside the projection  $\Pi$ . Baker and Stark proved that no further class-number-one field exists beyond  $d = 163$  [11, 12]; the QGT reads this as the closure of the generation sequence. The partition  $\mathcal{B}_{13} = S_5 \oplus T_8$  is the direct image of this geometric bifurcation.

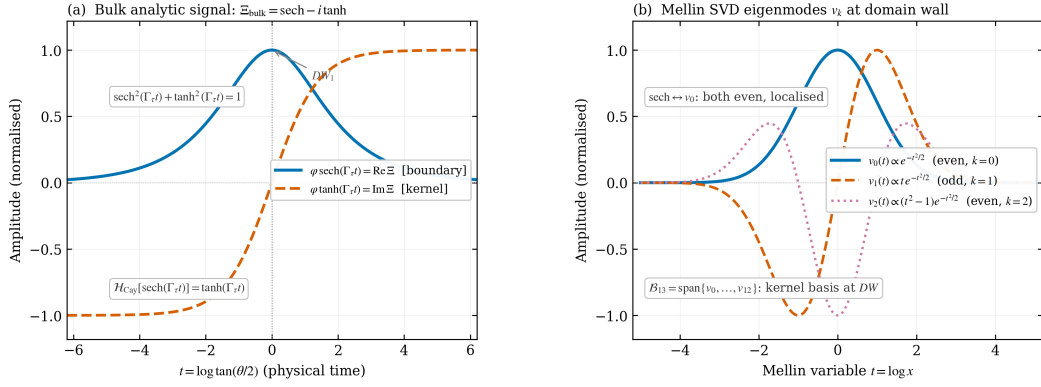


**Figure 2:** Analytic signal structure on  $S^1(\tau)$ . *Top:* the boundary component  $\xi(\theta)$  (real part, gravitational channel, blue) and kernel component  $\eta(\theta)$  (imaginary part, electromagnetic channel, red) oscillate in permanent quadrature; the envelope  $A(\theta)$  is their instantaneous amplitude. *Middle:* the Noether invariant  $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$  is conserved to numerical precision — it is the geometric mean of the analytic signal, maintained by the exchange equations  $\dot{\xi} = -\Gamma_\tau\eta$ ,  $\dot{\eta} = +\Gamma_\tau\xi$ . *Bottom:* the instantaneous phase  $\phi(\theta) = \Gamma_\tau\theta$  accumulates at the golden rate  $\Gamma_\tau = \varphi/\sqrt{5}$  but never closes ( $\Gamma_\tau \notin \mathbb{Q}$ ): this is the arrow of time.



**Figure 3:** Internal structure  $\mathcal{B}_{13} = S_5 \oplus T_8$  derived from the filter  $H(z) = P_4(z)/(1 - \alpha z)$ . The polynomial  $P_4$  generates the five golden scale modes ( $S_5$ ); the pole  $1/(1 - \alpha z)$  at the anisotropic lock-in generates the eight tensor degrees of freedom ( $T_8$ ), corresponding to the two chiral curvature channels and the three electroweak boson pairs. The total internal space  $\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$  has dimension  $27 = 1 + 13 + 13$ , with arithmetic index  $163 = 137 + 26$ .

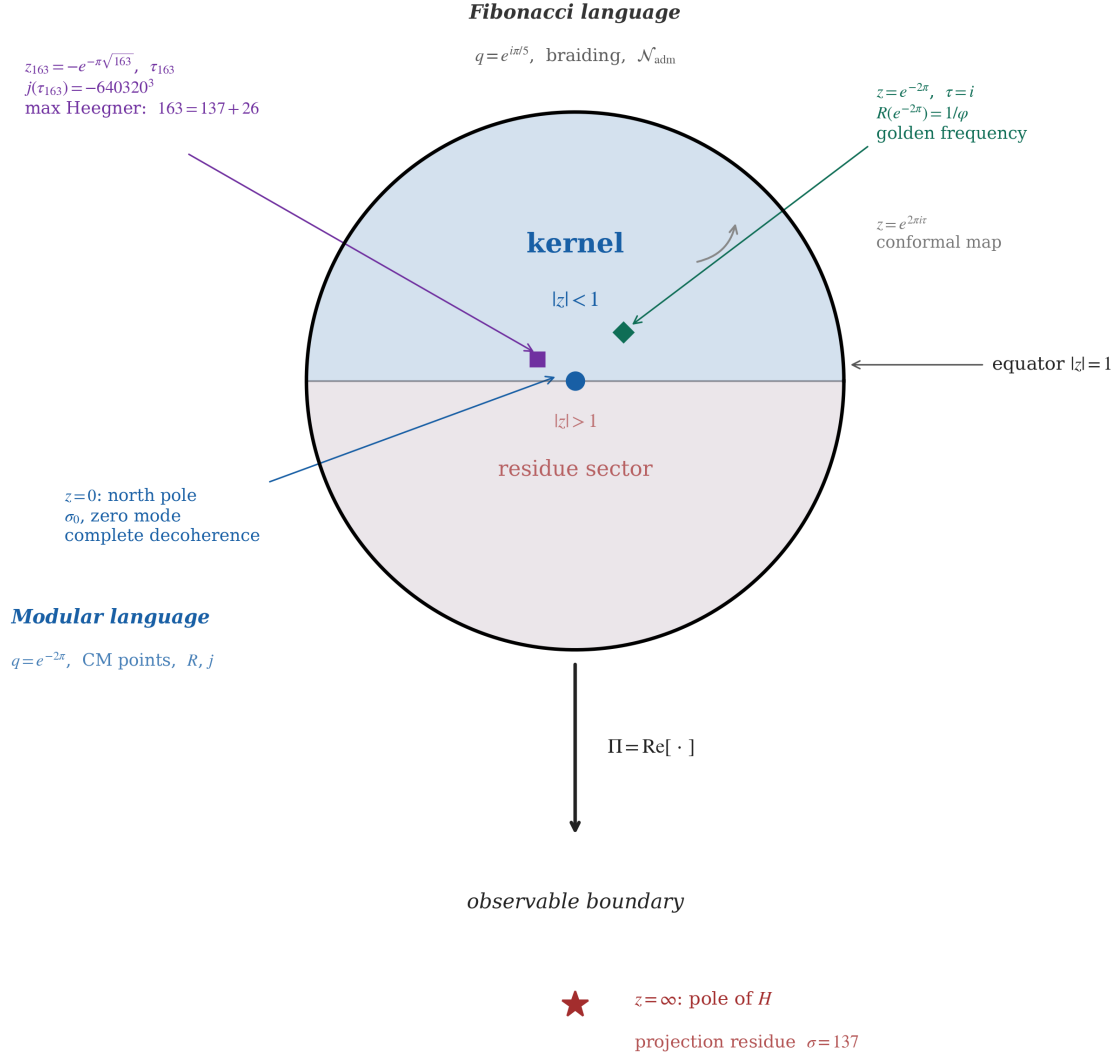




Both  $\Psi_*$  and  $v_0$  are even, localised, and decay monotonically.  $v_1$  and  $v_2$  carry odd/quadratic corrections. The structural equivalence  $\text{sech} \leftrightarrow v_0$  justifies  $\mathcal{B}_{13} = \text{span}\{v_0, \dots, v_{12}\}$  as the natural basis of  $\ker \Pi$  at the domain walls.

**Figure 4:** *Left:* the lock-in profile  $\Psi_*(n) = \varphi \text{sech}(n/\ell_{\text{pen}})$ , normalised. It is even, localised, flat at  $n = 0$  ( $\partial_n \Psi_*|_0 = 0$ ), and decays exponentially into the bulk. *Right:* the first three Mellin SVD eigenmodes  $v_k(t)$ ,  $t = \log x$ , with Gaussian envelope.  $v_0$  (even, localised, monotone decay) and the sech profile have the same structural form: both are the unique even, unit-norm, rapidly decaying function in their respective domains.  $v_1$  and  $v_2$  carry the odd and quadratic corrections. This structural equivalence  $\text{sech} \leftrightarrow v_0$  is the geometric justification for identifying  $\mathcal{B}_{13} = S_5 \oplus T_8$  as the span of the first 13 Mellin modes of the filtered kernel: the lock-in soliton and the Mellin basis inhabit the same function space, read from the normal and the logarithmic directions respectively.

**The filter  $H(z) = P_4(z)/(1 - \alpha z)$  on  $\mathbb{P}^1$**



**Figure 5:** The filter  $H(z) = P_4(z)/(1 - \alpha z)$  on the Riemann sphere  $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ . The unit circle (equator) is the observable boundary. The northern hemisphere ( $|z| < 1$ ) is the kernel. The north pole  $z = 0$  carries the zero mode  $\sigma_0$  (complete decoherence); the south pole  $z = \infty$  carries the projection residue  $\sigma = 137$ . The projection  $\Pi = \text{Re}[\cdot]$  extracts the real part on the equator. The conformal map  $z = e^{2\pi i\tau}$  continues the equator into the upper half-plane: the point  $z = e^{-2\pi}$  ( $\tau = i$ ) reads the golden frequency  $R(e^{-2\pi}) = 1/\phi$ ; the point  $z_{163} = -e^{-\pi\sqrt{163}}$  ( $\tau_{163}$ ) reads the maximum Heegner structure with  $163 = 137 + 26$ . The two languages — Fibonacci (equator) and modular (northern hemisphere) — are the same function read from two sides.